Model Architecture: gemma-4-E2B

Generated by sg_modelload

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1 Model Overview

Model name	gemma-4-E2B
Transformers class	`Gemma4ForConditionalGeneration`
Path	`/models/memorization/gemma-4-E2B`
Total parameters	5,104,297,504 (5.10B)
Weight dtype	bfloat16
Hidden size d	1,536
Decoder layers L	35
Query heads H	8
Key/value heads H_{kv}	1
Head dim d_h	256
MLP width d_{ff}	6,144
PLE dim per layer	256
KV-shared layers	20
Vocabulary V	262,144
Max context length	131,072
Sliding window w	512
Logit soft-cap	30.0
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	$\mathrm{GELU}_{\mathrm{tanh}}$
Vision encoder: layers	16
hidden size / heads	768 / 12
Audio encoder: layers	12
hidden size / heads	1024 / 8

How to read this document. Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Notation

B, T	batch size, sequence length (tokens)
d	hidden size (1536)
V	vocabulary size (262,144)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
$\mathrm{RMSNorm}(x)$	$\gamma \odot x/\sqrt{\mathrm{mean}(x^2)+\epsilon}$ (learned scale γ)
$\mathrm{LN}(x)$	LayerNorm: $\gamma \odot (x-\mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 The Model as a Function

Input: token ids $x \in \{0,\dots,V-1\}^{B\times T}$. Output: next-token logits $y \in \mathbb{R}^{B\times T\times V}$. The 35 decoder layers are written with composition powers, so a block that repeats n times appears once as $\mathcal{B}^{\circ n}$.

$$
\begin{aligned}
h_0 &= \mathrm{Merge}\Big(\mathrm{Embed}(x)\cdot\sqrt{d},\ \mathrm{Proj}_{\mathrm{img}}\big(\mathrm{Enc}_{\mathrm{V}}(\mathrm{img})\big),\ \mathrm{Proj}_{\mathrm{aud}}\big(\mathrm{Enc}_{\mathrm{A}}(\mathrm{aud})\big)\Big)\\
p &= \tfrac{1}{\sqrt{2}}\Big(\mathrm{Embed}_{\mathrm{PLE}}(x) + \mathrm{RMSNorm}\big(W_{\mathrm{ple}}\,h_0/\sqrt{d}\big)\Big),\quad p=(p_0,\dots,p_{L-1})\\
h_L &= \Big(\big(\mathcal{B}_{\mathrm{F2}}\circ\mathcal{B}_{\mathrm{S2}}^{\circ 4}\big)^{\circ 4}\circ\big(\mathcal{B}_{\mathrm{F1}}\circ\mathcal{B}_{\mathrm{S1}}^{\circ 4}\big)^{\circ 3}\Big)(h_0)\\
z &= W_E^{\top}\ \mathrm{RMSNorm}_{\mathrm{final}}(h_L)\\
y &= 30.0\cdot\tanh(z/30.0)
\end{aligned}
$$

Merge puts the projected image/audio vectors at the placeholder-token positions of the text embedding sequence. The encoders are themselves repeated stacks:

$$\mathrm{Enc_V} = \mathrm{Post} \circ \mathcal{B}_{\mathrm{V}}^{\circ 16} \circ \mathrm{PatchEmbed}$$
$$\mathrm{Enc_A} = \mathrm{Post} \circ \mathcal{B}_{\mathrm{A}}^{\circ 12} \circ \mathrm{Subsample}$$

Each decoder layer ℓ also receives its own per-layer embedding p_ℓ (written $\mathcal{B}(h)$ for brevity; strictly $\mathcal{B}(h, p_\ell)$).

2.1 Layer pattern

Symbol	Block type	Count	Params / block	Layer indices
$\mathcal{B}_{\mathrm{S1}}$	Sliding-window attention block (MLP width 6144)	12	36,184,064	0–3, 5–8, 10–13
$\mathcal{B}_{\mathrm{F1}}$	Full (global) attention block (MLP width 6144)	3	43,262,464	4, 9, …, 14 (every 5th)
$\mathcal{B}_{\mathrm{S2}}$	Sliding-window attention block (KV-shared, MLP width 12288)	16	63,708,928	15–18, 20–23, 25–28, 30–33
$\mathcal{B}_{\mathrm{F2}}$	Full (global) attention block (KV-shared, MLP width 12288)	4	70,000,640	19, 24, …, 34 (every 5th)
$\mathcal{B}_{\mathrm{V}}$	Vision encoder block	16	9,440,384	0–15
$\mathcal{B}_{\mathrm{A}}$	Conformer audio block	12	25,180,288	0–11

Language-model decoder (35 layers). layers 0–14: $[\mathcal{B}_{\mathrm{S1}}\times 4, \mathcal{B}_{\mathrm{F1}}]$ repeated 3×; layers 15–34: $[\mathcal{B}_{\mathrm{S2}}\times 4, \mathcal{B}_{\mathrm{F2}}]$ repeated 4×.

Vision encoder (16 layers). layers 0–15: $[\mathcal{B}_{\mathrm{V}}]$ repeated 16×.

Audio encoder (12 layers). layers 0–11: $[\mathcal{B}_{\mathrm{A}}]$ repeated 12×.

2.2 Block functions

$\mathcal{B}_{\mathrm{S1}}$ — **Sliding-window attention block (MLP width 6144).**

$$a = h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) \quad \text{norm before AND after attention}$$
$$b = a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) \quad \text{norm before AND after the MLP}$$
$$c = b + \mathrm{RMSNorm}_{\mathrm{ple}}\big(W_{\mathrm{proj}}(\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\, b) \odot p_\ell)\big) \quad \text{inject this layer's 256-dim per-layer embedding } p_\ell$$
$$\mathcal{B}(h) = \gamma_\ell \cdot c \quad \text{fixed per-layer scalar}$$
$$Q = W_Q\, u \quad H = 8 \text{ heads of size } d_h = 256$$
$$K,\ V = W_K\, u, \quad W_V\, u \quad H_{kv} = 1 \text{ key/value heads}$$
$$Q,\ K = \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) \quad \text{per-head normalisation}$$
$$V = \mathrm{RMSNorm}(V) \quad \text{no learned scale}$$
$$Q,\ K = \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) \quad \text{rotate by position}$$
$$S_h = s\, Q_h K_{g(h)}^{\top} + M \quad \text{mask } M\text{: causal + sliding window of } w = 512$$
$$P_h = \mathrm{softmax}(S_h) \quad \text{attention weights}$$
$$O_h = P_h\, V_{g(h)}$$
$$\mathrm{Attn}(u) = W_O\, [O_1; \ldots; O_H]$$
$$\mathrm{FFN}(u) = W_{\mathrm{down}}\big(\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\big) \quad \text{gated MLP, width 6144}$$

$\mathcal{B}_{\mathrm{F1}}$ — **Full (global) attention block (MLP width 6144).**

$$a = h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) \quad \text{norm before AND after attention}$$
$$b = a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) \quad \text{norm before AND after the MLP}$$
$$c = b + \mathrm{RMSNorm}_{\mathrm{ple}}\big(W_{\mathrm{proj}}(\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\, b) \odot p_\ell)\big) \quad \text{inject this layer's 256-dim per-layer embedding } p_\ell$$
$$\mathcal{B}(h) = \gamma_\ell \cdot c \quad \text{fixed per-layer scalar}$$
$$Q = W_Q\, u \quad H = 8 \text{ heads of size } d_h = 512$$

$$
\begin{aligned}
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 1 \text{ key/value heads} \\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation} \\
V &= \mathrm{RMSNorm}(V) && \text{no learned scale} \\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position} \\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal} \\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights} \\
O_h &= P_h\, V_{g(h)} \\
\mathrm{Attn}(u) &= W_O\, [O_1; \ldots; O_H] \\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\, \mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\big) && \text{gated MLP, width } 6144
\end{aligned}
$$

$\mathcal{B}_{\mathrm{S2}}$ — **Sliding-window attention block (KV-shared, MLP width 12288).**

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\, \mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention} \\
b &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\, \mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP} \\
c &= b + \mathrm{RMSNorm}_{\mathrm{ple}}\big(W_{\mathrm{proj}}(\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\, b) \odot p_\ell)\big) && \text{inject this layer's 256-dim per-layer embedding } p_\ell \\
\mathcal{B}(h) &= \gamma_\ell \cdot c && \text{fixed per-layer scalar} \\
Q &= W_Q\, u && H = 8 \text{ heads of size } d_h = 256 \\
K,\ V &= K^{(\ell^*)},\ V^{(\ell^*)} && \text{from last non-shared layer } \ell^* \text{ (no K/V weights)} \\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ K && \text{per-head normalisation} \\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position} \\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal + sliding window of } w = 512 \\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights} \\
O_h &= P_h\, V_{g(h)} \\
\mathrm{Attn}(u) &= W_O\, [O_1; \ldots; O_H] \\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\, \mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\big) && \text{gated MLP, width } 12288
\end{aligned}
$$

$\mathcal{B}_{\mathrm{F2}}$ — **Full (global) attention block (KV-shared, MLP width 12288).**

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\, \mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention} \\
b &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\, \mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP} \\
c &= b + \mathrm{RMSNorm}_{\mathrm{ple}}\big(W_{\mathrm{proj}}(\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\, b) \odot p_\ell)\big) && \text{inject this layer's 256-dim per-layer embedding } p_\ell \\
\mathcal{B}(h) &= \gamma_\ell \cdot c && \text{fixed per-layer scalar} \\
Q &= W_Q\, u && H = 8 \text{ heads of size } d_h = 512 \\
K,\ V &= K^{(\ell^*)},\ V^{(\ell^*)} && \text{from last non-shared layer } \ell^* \text{ (no K/V weights)} \\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ K && \text{per-head normalisation} \\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position} \\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal} \\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights} \\
O_h &= P_h\, V_{g(h)} \\
\mathrm{Attn}(u) &= W_O\, [O_1; \ldots; O_H] \\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\, \mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\big) && \text{gated MLP, width } 12288
\end{aligned}
$$

$\mathcal{B}_{\mathrm{V}}$ — **Vision encoder block.**

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\, \mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention} \\
\mathcal{B}(h) &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\, \mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP} \\
Q &= W_Q\, u && H = 12 \text{ heads of size } d_h = 64 \\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 12 \text{ key/value heads} \\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation} \\
V &= \mathrm{RMSNorm}(V) && \text{no learned scale} \\
Q,\ K &= \mathrm{RoPE}_{2D}(Q),\ \mathrm{RoPE}_{2D}(K) && \text{rotate by (row, column) patch position}
\end{aligned}
$$

$$
\begin{aligned}
S_h &= s\, Q_h K_h^\top + M && \text{mask } M\text{: bidirectional}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_h\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H]\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 3072}
\end{aligned}
$$

$\mathcal{B}_{\mathrm{A}}$ — **Conformer audio block.**

$$
\begin{aligned}
h_1 &= h + \tfrac{1}{2}\ \mathrm{FFN}_1(h) && \text{macaron half-step FFN}\\
h_2 &= h_1 + \mathrm{RMSNorm}_{\mathrm{post}}\big(\,\mathrm{Attn}_{rel}(\mathrm{RMSNorm}_{\mathrm{pre}}(h_1))\big) && \text{local attention, relative positions}\\
h_3 &= h_2 + W_{\mathrm{end}}\ \mathrm{SiLU}\big(\,\mathrm{RMSNorm}(\mathrm{DWConv}(\mathrm{GLU}(W_{\mathrm{start}}\,\mathrm{RMSNorm}(h_2))))\big) && \text{convolution module}\\
h_4 &= h_3 + \tfrac{1}{2}\ \mathrm{FFN}_2(h_3)\\
\mathcal{B}(h) &= \mathrm{RMSNorm}_{\mathrm{out}}(h_4)\\
\mathrm{FFN}_i(u) &= \mathrm{RMSNorm}\big(W_2\ \mathrm{SiLU}(W_1\ \mathrm{RMSNorm}(u))\big)
\end{aligned}
$$

3 Forward Pass in Pseudocode

Algorithm 1 gemma-4-E2B: full forward pass

Require: token ids x of shape $B \times T$; optional images / audio

$h \leftarrow \mathrm{Embed}(x) \cdot \sqrt{d}$ ▷ look up one vector per token: $B{\times}T{\times}1536$
if image input present **then**
  $e \leftarrow \mathrm{Enc}_{\mathrm{V}}(\text{image})$ ▷ vision encoder: 16 layers, Algorithm 2
  $h[\text{image token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 1536
end if
if audio input present **then**
  $e \leftarrow \mathrm{Enc}_{\mathrm{A}}(\text{audio})$ ▷ audio encoder: 12 layers, Algorithm 3
  $h[\text{audio token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 1536
end if
$p \leftarrow \mathrm{PerLayerInputs}(x, h)$ ▷ 35 small vectors per token, one for each layer
for $r \leftarrow 0$ **to** 2 **do** ▷ layers 0–14: pattern of 5 repeated 3 times
  for $j \leftarrow 1$ **to** 4 **do**
    $h \leftarrow \mathcal{B}_{\mathrm{S1}}(h, p_\ell)$ ▷ Sliding-window attention block (MLP width 6144)
  end for
  $h \leftarrow \mathcal{B}_{\mathrm{F1}}(h, p_\ell)$ ▷ Full (global) attention block (MLP width 6144)
end for
for $r \leftarrow 0$ **to** 3 **do** ▷ layers 15–34: pattern of 5 repeated 4 times
  for $j \leftarrow 1$ **to** 4 **do**
    $h \leftarrow \mathcal{B}_{\mathrm{S2}}(h, p_\ell)$ ▷ Sliding-window attention block (KV-shared, MLP width 12288)
  end for
  $h \leftarrow \mathcal{B}_{\mathrm{F2}}(h, p_\ell)$ ▷ Full (global) attention block (KV-shared, MLP width 12288)
end for
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$
$y \leftarrow h\, W_E^\top$ ▷ logits $B{\times}T{\times}262,144$, embedding matrix re-used
$y \leftarrow 30.0 \tanh(y/30.0)$ ▷ soft-cap logits
return y ▷ $\mathrm{softmax}(y)$ = next-token probabilities

Algorithm 2 Vision encoder

$h \leftarrow \mathrm{PatchEmbed}(\text{pixels}) + \text{position embedding}$ ▷ one vector per image patch
for $\ell \leftarrow 0$ **to** 15 **do**
  $h \leftarrow \mathcal{B}_{\mathrm{V}}(h)$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 2
return h

Algorithm 3 Audio encoder

$h \leftarrow$ Subsample(mel spectrogram) ▷ 2 strided convs: 4× fewer frames
for $\ell \leftarrow 0$ **to** 11 **do**
    $h \leftarrow \mathcal{B}_{\mathrm{A}}(h)$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 2
return h

Algorithm 4 Block $\mathcal{B}_{\mathrm{S1}}(h)$ — Sliding-window attention block (MLP width 6144)

Require: h of shape $B \times T \times 1536$, per-layer input p_ℓ

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}1{\times}256$ (1 key/value heads, each shared by 8 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$V \leftarrow \mathrm{RMSNorm}(V)$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal + only the last 512 tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}256$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1536$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 6144, gate, shrink back to 1536
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
per-layer embedding injection
$h \leftarrow h + \mathrm{RMSNorm}\left((\mathrm{GELU}_{\tanh}(hW_{\mathrm{gate}}^\top) \odot p_\ell)W_{\mathrm{proj}}^\top\right)$ ▷ gate down to 256, multiply by p_ℓ, project back up
$h \leftarrow \gamma_\ell\, h$ ▷ fixed per-layer scale
return h

Algorithm 5 Block $\mathcal{B}_{\mathrm{F1}}(h)$ — Full (global) attention block (MLP width 6144)

Require: h of shape $B \times T \times 1536$, per-layer input p_ℓ

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}512$ (8 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}1{\times}512$ (1 key/value heads, each shared by 8 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$V \leftarrow \mathrm{RMSNorm}(V)$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}512$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1536$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 6144, gate, shrink back to 1536
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
per-layer embedding injection
$h \leftarrow h + \mathrm{RMSNorm}\left((\mathrm{GELU}_{\tanh}(hW_{\mathrm{gate}}^\top) \odot p_\ell)W_{\mathrm{proj}}^\top\right)$ ▷ gate down to 256, multiply by p_ℓ, project back up
$h \leftarrow \gamma_\ell\, h$ ▷ fixed per-layer scale
return h

Algorithm 6 Block $\mathcal{B}_{\mathrm{S2}}(h)$ — Sliding-window attention block (KV-shared, MLP width 12288)

Require: h of shape $B \times T \times 1536$, per-layer input p_ℓ

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 query heads)
$K, V \leftarrow K^{(\ell^*)}, V^{(\ell^*)}$ ▷ re-use keys/values of an earlier layer (saves compute and KV-cache)
$Q \leftarrow \mathrm{RMSNorm}(Q)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal + only the last 512 tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}256$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1536$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 12288, gate, shrink back to 1536
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
per-layer embedding injection
$h \leftarrow h + \mathrm{RMSNorm}\left((\mathrm{GELU}_{\tanh}(hW_{\mathrm{gate}}^\top) \odot p_\ell)W_{\mathrm{proj}}^\top\right)$ ▷ gate down to 256, multiply by p_ℓ, project back up
$h \leftarrow \gamma_\ell\, h$ ▷ fixed per-layer scale
return h

Algorithm 7 Block $\mathcal{B}_{\mathrm{F2}}(h)$ — Full (global) attention block (KV-shared, MLP width 12288)

Require: h of shape $B \times T \times 1536$, per-layer input p_ℓ

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}512$ (8 query heads)
$K, V \leftarrow K^{(\ell^*)}, V^{(\ell^*)}$ ▷ re-use keys/values of an earlier layer (saves compute and KV-cache)
$Q \leftarrow \mathrm{RMSNorm}(Q)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}512$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1536$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 12288, gate, shrink back to 1536
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
per-layer embedding injection
$h \leftarrow h + \mathrm{RMSNorm}\left((\mathrm{GELU}_{\tanh}(hW_{\mathrm{gate}}^\top) \odot p_\ell)W_{\mathrm{proj}}^\top\right)$ ▷ gate down to 256, multiply by p_ℓ, project back up
$h \leftarrow \gamma_\ell\, h$ ▷ fixed per-layer scale
return h

Algorithm 8 Block $\mathcal{B}_{\mathrm{V}}(h)$ — Vision encoder block

Require: h of shape $B \times T \times 768$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}12{\times}64$ (12 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}12{\times}64$ (12 key/value heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$V \leftarrow \mathrm{RMSNorm}(V)$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ 2-D rotary position (row, column)
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: every patch/frame sees every other
$O \leftarrow PV$ ▷ $B{\times}T{\times}12{\times}64$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}768$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 3072, gate, shrink back to 768
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
return h

Algorithm 9 Block $\mathcal{B}_{\mathrm{A}}(h)$ — Conformer audio block

Require: h of shape $B \times T \times 1024$

$h \leftarrow h + \frac{1}{2}\,\mathrm{FFN}_1(h)$ ▷ first half-step feed-forward
$h \leftarrow h + \mathrm{RMSNorm}(\mathrm{Attn}_{rel}(\mathrm{RMSNorm}(h)))$ ▷ local self-attention over audio frames
$h \leftarrow h + \mathrm{ConvModule}(h)$ ▷ GLU → depthwise conv → norm → SiLU → linear
$h \leftarrow h + \frac{1}{2}\,\mathrm{FFN}_2(h)$ ▷ second half-step feed-forward
$h \leftarrow \mathrm{RMSNorm}(h)$
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

Weight	Type	Shape	Params
`model.vision_tower.patch_embedder.position_embedding_table`	Gemma4VisionPatchEmbedder	$2 \times 10240 \times 768$	15,728,640
`model.vision_tower.patch_embedder.input_proj.weight`	Linear	768×768	589,824
`model.language_model.embed_tokens.weight`	Gemma4TextScaledWordEmbedding	262144×1536	402,653,184
`model.language_model.norm.weight`	Gemma4RMSNorm	1536	1,536
`model.language_model.embed_tokens_per_layer.weight`	Gemma4TextScaledWordEmbedding	262144×8960	2,348,810,240
`model.language_model.per_layer_model_projection.weight`	Linear	8960×1536	13,762,560
`model.language_model.per_layer_projection_norm.weight`	Gemma4RMSNorm	256	256
`model.audio_tower.subsample_conv_projection.layer0.conv.weight`	Conv2d	$128 \times 1 \times 3 \times 3$	1,152
`model.audio_tower.subsample_conv_projection.layer0.norm.weight`	LayerNorm	128	128

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Weight	Type	Shape	Params
`model.audio_tower.subsample_conv_projection.layer1.conv.weight`	Conv2d	$32 \times 128 \times 3 \times 3$	36,864
`model.audio_tower.subsample_conv_projection.layer1.norm.weight`	LayerNorm	32	32
`model.audio_tower.subsample_conv_projection.input_proj_linear.weight`	Linear	1024×1024	1,048,576
`model.audio_tower.output_proj.weight`	Linear	1536×1024	1,572,864
`model.audio_tower.output_proj.bias`	Linear	1536	1,536
`model.embed_vision.embedding_projection.weight`	Linear	1536×768	1,179,648
`model.embed_audio.embedding_projection.weight`	Linear	1536×1536	2,359,296
`lm_head.weight` (tied to `model.language_model.embed_tokens.weight`)	Linear	262144×1536	0 (shared)

4.2 Language-model decoder: `model.language_model.layers`

Table 3: $\mathcal{B}_{\mathrm{S1}}$: Sliding-window attention block (MLP width 6144) — layers 0–3, 5–8, 10–13; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	2048×1536	3,145,728
`self_attn.q_norm.weight`	Gemma4RMSNorm	256	256
`self_attn.k_norm.weight`	Gemma4RMSNorm	256	256
`self_attn.k_proj.weight`	Linear	256×1536	393,216
`self_attn.v_proj.weight`	Linear	256×1536	393,216
`self_attn.o_proj.weight`	Linear	1536×2048	3,145,728
`mlp.gate_proj.weight`	Linear	6144×1536	9,437,184
`mlp.up_proj.weight`	Linear	6144×1536	9,437,184
`mlp.down_proj.weight`	Linear	1536×6144	9,437,184
`input_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_attention_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`per_layer_input_gate.weight`	Linear	256×1536	393,216
`per_layer_projection.weight`	Linear	1536×256	393,216
`post_per_layer_input_norm.weight`	Gemma4RMSNorm	1536	1,536
Per block			**36,184,064**
× 12 blocks			**434,208,768**

Table 4: $\mathcal{B}_{\mathrm{F1}}$: Full (global) attention block (MLP width 6144) — layers 4, 9, …, 14 (every 5th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×1536	6,291,456
`self_attn.q_norm.weight`	Gemma4RMSNorm	512	512
`self_attn.k_norm.weight`	Gemma4RMSNorm	512	512
`self_attn.k_proj.weight`	Linear	512×1536	786,432
`self_attn.v_proj.weight`	Linear	512×1536	786,432
`self_attn.o_proj.weight`	Linear	1536×4096	6,291,456
`mlp.gate_proj.weight`	Linear	6144×1536	9,437,184

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Weight (relative)	Type	Shape	Params
`mlp.up_proj.weight`	Linear	6144×1536	9,437,184
`mlp.down_proj.weight`	Linear	1536×6144	9,437,184
`input_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_attention_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`per_layer_input_gate.weight`	Linear	256×1536	393,216
`per_layer_projection.weight`	Linear	1536×256	393,216
`post_per_layer_input_norm.weight`	Gemma4RMSNorm	1536	1,536
Per block			**43,262,464**
× 3 blocks			**129,787,392**

Table 5: $\mathcal{B}_{S2}$: Sliding-window attention block (KV-shared, MLP width 12288) — layers 15–18, 20–23, 25–28, 30–33; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	2048×1536	3,145,728
`self_attn.q_norm.weight`	Gemma4RMSNorm	256	256
`self_attn.o_proj.weight`	Linear	1536×2048	3,145,728
`mlp.gate_proj.weight`	Linear	12288×1536	18,874,368
`mlp.up_proj.weight`	Linear	12288×1536	18,874,368
`mlp.down_proj.weight`	Linear	1536×12288	18,874,368
`input_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_attention_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`per_layer_input_gate.weight`	Linear	256×1536	393,216
`per_layer_projection.weight`	Linear	1536×256	393,216
`post_per_layer_input_norm.weight`	Gemma4RMSNorm	1536	1,536
Per block			**63,708,928**
× 16 blocks			**1,019,342,848**

Table 6: $\mathcal{B}_{F2}$: Full (global) attention block (KV-shared, MLP width 12288) — layers 19, 24, …, 34 (every 5th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×1536	6,291,456
`self_attn.q_norm.weight`	Gemma4RMSNorm	512	512
`self_attn.o_proj.weight`	Linear	1536×4096	6,291,456
`mlp.gate_proj.weight`	Linear	12288×1536	18,874,368
`mlp.up_proj.weight`	Linear	12288×1536	18,874,368
`mlp.down_proj.weight`	Linear	1536×12288	18,874,368
`input_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_attention_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`per_layer_input_gate.weight`	Linear	256×1536	393,216
`per_layer_projection.weight`	Linear	1536×256	393,216
`post_per_layer_input_norm.weight`	Gemma4RMSNorm	1536	1,536
Per block			**70,000,640**
× 4 blocks			**280,002,560**

4.3 Vision encoder: `model.vision_tower.encoder.layers`

Table 7: $\mathcal{B}_{\mathrm{V}}$: Vision encoder block — layers 0–15; weight names are relative to `model.vision_tower.encoder.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.linear.weight`	Linear	768×768	589,824
`self_attn.k_proj.linear.weight`	Linear	768×768	589,824
`self_attn.v_proj.linear.weight`	Linear	768×768	589,824
`self_attn.o_proj.linear.weight`	Linear	768×768	589,824
`self_attn.q_norm.weight`	Gemma4RMSNorm	64	64
`self_attn.k_norm.weight`	Gemma4RMSNorm	64	64
`mlp.gate_proj.linear.weight`	Linear	3072×768	2,359,296
`mlp.up_proj.linear.weight`	Linear	3072×768	2,359,296
`mlp.down_proj.linear.weight`	Linear	768×3072	2,359,296
`input_layernorm.weight`	Gemma4RMSNorm	768	768
`post_attention_layernorm.weight`	Gemma4RMSNorm	768	768
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	768	768
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	768	768
Per block			**9,440,384**
$\times$ 16 blocks			**151,046,144**

4.4 Audio encoder: `model.audio_tower.layers`

Table 8: $\mathcal{B}_{\mathrm{A}}$: Conformer audio block — layers 0–11; weight names are relative to `model.audio_tower.layers.`ℓ

Weight (relative)	Type	Shape	Params
`feed_forward1.ffw_layer_1.linear.weight`	Linear	4096×1024	4,194,304
`feed_forward1.ffw_layer_2.linear.weight`	Linear	1024×4096	4,194,304
`feed_forward1.pre_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`feed_forward1.post_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`feed_forward2.ffw_layer_1.linear.weight`	Linear	4096×1024	4,194,304
`feed_forward2.ffw_layer_2.linear.weight`	Linear	1024×4096	4,194,304
`feed_forward2.pre_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`feed_forward2.post_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`self_attn.per_dim_scale`	Gemma4AudioAttention	128	128
`self_attn.q_proj.linear.weight`	Linear	1024×1024	1,048,576
`self_attn.k_proj.linear.weight`	Linear	1024×1024	1,048,576
`self_attn.v_proj.linear.weight`	Linear	1024×1024	1,048,576
`self_attn.post.linear.weight`	Linear	1024×1024	1,048,576
`self_attn.relative_k_proj.weight`	Linear	1024×1024	1,048,576
`lconv1d.linear_start.linear.weight`	Linear	2048×1024	2,097,152
`lconv1d.linear_end.linear.weight`	Linear	1024×1024	1,048,576
`lconv1d.depthwise_conv1d.weight`	Gemma4AudioCausalConv1d	$1024 \times 1 \times 5$	5,120
`lconv1d.pre_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`lconv1d.conv_norm.weight`	Gemma4RMSNorm	1024	1,024
`norm_pre_attn.weight`	Gemma4RMSNorm	1024	1,024
`norm_post_attn.weight`	Gemma4RMSNorm	1024	1,024
`norm_out.weight`	Gemma4RMSNorm	1024	1,024
Per block			**25,180,288**

continued on next page

Weight (relative)	Type	Shape	Params
× **12 blocks**			**302,163,456**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Per-layer token embedding (PLE) `embed_tokens_per_layer`	A second embedding table looked up from the same token ids. It holds one small vector per token for every decoder layer, which gives each layer extra token-specific information cheaply. It can sit in CPU/flash memory, which is what the "E" (effective size) in the model name refers to.
PLE gate (down-projection) `layers.per_layer_input_gate`	Inside each layer: projects the hidden state down to the PLE size; after GELU it multiplies this layer's per-layer embedding p_ℓ.
PLE up-projection `layers.per_layer_projection`	Projects the gated per-layer signal back up to the hidden size before it is added to the residual stream.
PLE output norm `layers.post_per_layer_input_norm`	RMSNorm on the per-layer contribution before the residual add.
PLE context projection `per_layer_model_projection`	Projects the main token embedding into one small vector per layer. Added to the PLE lookup, so the per-layer input depends on context as well as token identity.
PLE projection norm `per_layer_projection_norm`	RMSNorm on the projected per-layer inputs.
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	RMSNorm applied to the attention output before it is added to the residual stream (Gemma "sandwich" norm).
Pre-MLP norm (pre_feedforward_layernorm) `layers.pre_feedforward_layernorm`	RMSNorm on the residual stream before the MLP (Gemma).

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Layer (appears as)	What it does
Post-MLP norm (post_feedforward_layernorm) `layers.post_feedforward_layernorm`	RMSNorm on the MLP output before it is added back (Gemma "sandwich" norm).
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.
VISION ENCODER	
2-D position table `vision_tower.patch_embedder`	Learned vectors for a patch's x and y coordinate, added so the encoder knows where each patch sits in the image.
Patch embedding `patch_embedder.input_proj`	Each image patch (16×16 pixels × 3 colours, flattened) is mapped linearly to one vector.
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	RMSNorm applied to the attention output before it is added to the residual stream (Gemma "sandwich" norm).
Pre-MLP norm (pre_feedforward_layernorm) `layers.pre_feedforward_layernorm`	RMSNorm on the residual stream before the MLP (Gemma).
Post-MLP norm (post_feedforward_layernorm) `layers.post_feedforward_layernorm`	RMSNorm on the MLP output before it is added back (Gemma "sandwich" norm).
AUDIO ENCODER	
Subsampling convolution `layer0.conv`, `layer1.conv`	2-D convolution (stride 2) over the time × mel-frequency spectrogram. Two of them shorten the time axis 4×, so the encoder works on fewer, richer frames.
Subsampling norm `layer0.norm`, `layer1.norm`	LayerNorm over channels after each subsampling convolution.
Audio input projection `subsample_conv_projection.input_proj_linear`	Flattens each frame's conv features and projects them to the audio-encoder width.
Conformer FFN up-projection `feed_forward1.ffw_layer_1`, `feed_forward2.ffw_layer_1`	Expands each frame vector (e.g. 1024→4096) before the SiLU non-linearity.
Conformer FFN down-projection `feed_forward1.ffw_layer_2`, `feed_forward2.ffw_layer_2`	Projects back to encoder width. The result is halved ("macaron" half-step) and added to the residual.

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Layer (appears as)	What it does
Conformer FFN pre-norm `feed_forward1.pre_layer_norm`, `feed_forward2.pre_layer_norm`	RMSNorm before each of the two conformer feed-forward modules.
Conformer FFN post-norm `feed_forward1.post_layer_norm`, `feed_forward2.post_layer_norm`	RMSNorm on the feed-forward output before the residual add.
Per-dimension query scale `layers.self_attn`	A learned scale per head dimension of the query, replacing the fixed $1/\sqrt{d_h}$.
Audio Q/K/V projections `self_attn.q_proj`, `self_attn.k_proj`, `self_attn.v_proj`	Turn each audio frame into a query (what am I looking for), key (what do I contain) and value (what do I pass on). "Clippable" linears clamp activations for numerical safety.
Audio attention output projection `self_attn.post`	Mixes the concatenated head outputs back to encoder width.
Relative-position key projection `self_attn.relative_k_proj`	Projects relative-distance encodings so attention scores depend on how far apart two frames are (Transformer-XL style) instead of RoPE.
Conv-module input projection (GLU) `lconv1d.linear_start`	Doubles the width; a Gated Linear Unit then uses one half of the channels to gate the other half.
Conv-module output projection `lconv1d.linear_end`	Projects the convolved features back before the residual add.
Depthwise causal 1-D convolution `lconv1d.depthwise_conv1d`	Each channel is convolved over time with its own small kernel (5 frames), picking up local acoustic patterns such as phoneme transitions.
Conv-module pre-norm `lconv1d.pre_layer_norm`	RMSNorm before the light-weight convolution module.
Conv-module norm `lconv1d.conv_norm`	RMSNorm after the depthwise convolution.
Pre-attention norm `layers.norm_pre_attn`	RMSNorm applied to frames before audio self-attention.
Post-attention norm `layers.norm_post_attn`	RMSNorm on the attention output before the residual add.
Block output norm `layers.norm_out`	RMSNorm at the end of every conformer block.
Audio output projection `audio_tower.output_proj`	Maps the last audio-encoder states to the size the audio embedder expects.
MULTIMODAL PROJECTOR	
Image-to-text projector `embed_vision.embedding_projection`	Normalises pooled vision features and projects them into the language model's hidden size ("soft image tokens").
Audio-to-text projector `embed_audio.embedding_projection`	Normalises audio-encoder outputs and projects them into the language model's hidden size, so they can take the place of text-token embeddings.

6 Parameter Count

Table 10: Parameters by component

Component	Parameters	% of total
Per-layer embeddings (PLE)	2,390,151,936	46.8%
MLP / MoE	1,557,135,360	30.5%
Token embedding	402,653,184	7.9%
Audio encoder	304,824,608	6.0%
Attention / token mixer	278,412,288	5.5%
Vision encoder	167,364,608	3.3%

continued on next page

Component	Parameters	% of total
Multimodal projector	3,538,944	0.1%
Norms	216,576	0.0%
Total	**5,104,297,504**	**100.0%**

Table 11: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{S1}$ Sliding-window attention block (MLP width 6144)	12	36,184,064	434,208,768
$\mathcal{B}_{F1}$ Full (global) attention block (MLP width 6144)	3	43,262,464	129,787,392
$\mathcal{B}_{S2}$ Sliding-window attention block (KV-shared, MLP width 12288)	16	63,708,928	1,019,342,848
$\mathcal{B}_{F2}$ Full (global) attention block (KV-shared, MLP width 12288)	4	70,000,640	280,002,560
$\mathcal{B}_{V}$ Vision encoder block	16	9,440,384	151,046,144
$\mathcal{B}_{A}$ Conformer audio block	12	25,180,288	302,163,456
Modules outside stacks	1	–	2,787,746,336
Total			**5,104,297,504**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	262144 × 1536	bfloat16	402,653,184
`model`	Gemma4Model	–	–	–
`model.audio_tower`	Gemma4AudioModel	–	–	–
`model.audio_tower.layers.0`	Gemma4AudioLayer	–	–	–
`model.audio_tower.layers.0.feed_forward1`	Gemma4AudioFeed Forward	–	–	–
`model.audio_tower.layers.0.feed_forward1. ffw_layer_1`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.feed_forward1. ffw_layer_1.linear.weight`	Linear	4096 × 1024	bfloat16	4,194,304
`model.audio_tower.layers.0.feed_forward1. ffw_layer_2`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.feed_forward1. ffw_layer_2.linear.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.audio_tower.layers.0.feed_forward1. post_layer_norm.weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.feed_forward1. pre_layer_norm.weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.feed_forward2`	Gemma4AudioFeed Forward	–	–	–
`model.audio_tower.layers.0.feed_forward2. ffw_layer_1`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.feed_forward2. ffw_layer_1.linear.weight`	Linear	4096 × 1024	bfloat16	4,194,304
`model.audio_tower.layers.0.feed_forward2. ffw_layer_2`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.feed_forward2. ffw_layer_2.linear.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.audio_tower.layers.0.feed_forward2. post_layer_norm.weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.feed_forward2. pre_layer_norm.weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.lconv1d`	Gemma4AudioLight Conv1d	–	–	–
`model.audio_tower.layers.0.lconv1d.conv_norm. weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.lconv1d. depthwise_conv1d.weight`	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
`model.audio_tower.layers.0.lconv1d.linear_end`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.lconv1d. linear_end.linear.weight`	Linear	1024 × 1024	bfloat16	1,048,576

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.0.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.0.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.0.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.0.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.0.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.0.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.0.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.0.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.0.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.0.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.0.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.0.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.1.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.1.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.1.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.1.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.1.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.1.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.1.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.1.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.1.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.1.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.1.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.1.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.1.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.2.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.2.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.2.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.2.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.2.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.2.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.2.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.2.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.2.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.2.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.2.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.2.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.2.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.3.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.3.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.3.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.3.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.3.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.3.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.3.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.3.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.3.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.3.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.3.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.3.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.3.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.4.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.4.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.4.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.4.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.4.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.4.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.4.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.4.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.4.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.4.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.4.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.4.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.5.feed_forward1	Gemma4AudioFeedForward	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.5.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.5.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.5.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.5.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.5.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.5.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.5.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.5.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.5.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.5.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.5.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.6.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.6.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.6.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.6.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.6.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.6.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.6.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.6.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.6.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.6.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.6.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.6.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.7.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.7.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.7.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.7.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.7.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.7.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.7.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.7.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.7.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.7.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.7.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.7.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.7.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.8.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.8.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.8.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.8.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.8.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.8.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.8.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.8.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.8.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.8.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.8.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.8.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.8.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.9.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.9.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.9.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.9.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.9.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.9.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.9.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.9.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.9.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.9.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.9.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.9.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.10.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.10.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.10.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.10.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.10.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.10.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.10.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.10.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.10.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.10.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.10.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.10.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.11.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.11.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.11.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.11.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.11.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.11.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.11.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.11.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.11.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.11.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.11.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.11.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.11.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.output_proj.bias	Linear	1536	bfloat16	1,536
model.audio_tower.output_proj.weight	Linear	1536 × 1024	bfloat16	1,572,864
model.audio_tower.subsample_conv_projection	Gemma4AudioSubSampleConvProjection	–	–	–
model.audio_tower.subsample_conv_projection.input_proj_linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.subsample_conv_projection.layer0	Gemma4AudioSubSampleConvProjectionLayer	–	–	–
model.audio_tower.subsample_conv_projection.layer0.conv.weight	Conv2d	128 × 1 × 3 × 3	bfloat16	1,152
model.audio_tower.subsample_conv_projection.layer0.norm.weight	LayerNorm	128	bfloat16	128
model.audio_tower.subsample_conv_projection.layer1	Gemma4AudioSubSampleConvProjectionLayer	–	–	–
model.audio_tower.subsample_conv_projection.layer1.conv.weight	Conv2d	32 × 128 × 3 × 3	bfloat16	36,864
model.audio_tower.subsample_conv_projection.layer1.norm.weight	LayerNorm	32	bfloat16	32
model.embed_audio	Gemma4MultimodalEmbedder	–	–	–
model.embed_audio.embedding_projection.weight	Linear	1536 × 1536	bfloat16	2,359,296
model.embed_vision	Gemma4MultimodalEmbedder	–	–	–
model.embed_vision.embedding_projection.weight	Linear	1536 × 768	bfloat16	1,179,648
model.language_model	Gemma4TextModel	–	–	–
model.language_model.embed_tokens.weight	Gemma4TextScaledWordEmbedding	262144 × 1536	bfloat16	402,653,184
model.language_model.embed_tokens_per_layer.weight	Gemma4TextScaledWordEmbedding	262144 × 8960	bfloat16	2,348,810,240
model.language_model.layers.0	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.0.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.0.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.0.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.0.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.0.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.0.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.0.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.self_attn	Gemma4TextAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.0.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.0.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.0.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.0.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.0.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.0.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.1	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.1.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.1.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.1.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.1.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.1.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.1.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.1.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.1.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.1.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.1.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.1.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.1.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.1.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.2	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.2.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.2.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.2.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.2.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.2.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.2.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.2.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.2.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.2.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.2.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.2.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.2.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.3	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.3.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.3.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.3.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.3.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.3.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.3.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.3.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.3.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.3.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.3.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.4	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.4.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.4.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.4.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.4.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.4.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.4.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.4.self_attn.k_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.4.self_attn.k_proj.weight	Linear	512 × 1536	bfloat16	786,432

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.4.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.4.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.4.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.4.self_attn.v_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.5	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.5.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.5.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.5.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.5.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.5.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.5.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.5.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.5.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.5.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.5.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.5.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.5.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.6	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.6.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.6.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.6.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.6.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.6.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.6.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.6.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.6.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.6.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.6.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.6.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.6.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.7	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.7.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.7.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.7.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.7.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.7.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.7.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.7.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.7.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.7.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.7.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.8	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.8.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.8.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.8.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.8.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.8.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.8.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.8.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.8.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.8.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.8.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.8.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.8.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.9	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.9.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.9.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.9.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.9.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.9.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.9.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.9.self_attn.k_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.9.self_attn.k_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.9.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.9.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.9.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.9.self_attn.v_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.10	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.10.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.10.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.10.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.10.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.10.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.10.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.10.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.10.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.10.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.10.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.10.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.10.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.11	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.11.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.11.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.11.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.11.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.11.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.11.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.11.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.11.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.11.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.11.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.12	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.12.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.12.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.12.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.12.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.12.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.12.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.12.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.12.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.12.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.12.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.12.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.12.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.13	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.13.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.13.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.13.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.13.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.13.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.13.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.13.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.13.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.13.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.13.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.13.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.13.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.14	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.14.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.14.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.14.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.14.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.14.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.14.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.14.self_attn.k_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.14.self_attn.k_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.14.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.14.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.14.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.14.self_attn.v_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.15	Gemma4TextDecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.15.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.15.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.15.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.15.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.15.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.15.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.15.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.15.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.16	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.16.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.16.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.16.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.16.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.16.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.16.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.16.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.16.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.16.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.17	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.17.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.17.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.17.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.17.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.17.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.17.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.17.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.17.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.17.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.18	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.18.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.18.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.18.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.18.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.18.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.18.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.18.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.18.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.18.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.19	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.19.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.19.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.19.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.19.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.19.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.19.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.19.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.19.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.19.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.20	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.20.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.20.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.20.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.20.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.20.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.20.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.20.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.20.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.20.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.21	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.21.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.21.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.21.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.21.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.21.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.21.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.21.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.21.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.21.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.22	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.22.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.22.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.22.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.22.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.22.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.22.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.22.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.22.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.22.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.23	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.23.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.23.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.23.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.23.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.23.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.23.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.23.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.23.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.24	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.24.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.24.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.24.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.24.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.24.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.24.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.24.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.self_attn	Gemma4TextAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.24.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.24.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.24.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.25	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.25.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.25.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.25.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.25.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.25.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.25.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.25.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.25.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.25.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.25.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.26	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.26.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.26.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.26.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.26.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.26.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.26.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.26.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.26.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.26.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.26.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.27	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.27.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.27.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.27.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.27.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.27.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.27.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.27.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.27.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.27.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.28	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.28.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.28.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.28.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.28.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.28.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.28.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.28.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.28.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.28.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.28.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.29	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.29.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.29.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.29.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.29.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.29.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.29.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.29.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.29.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.29.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.29.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.29.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.29.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.29.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.29.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.29.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.30	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.30.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.30.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.30.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.30.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.30.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.30.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.30.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.30.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.30.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.30.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.31	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.31.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.31.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.31.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.31.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.31.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.31.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.31.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.31.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.31.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.32	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.32.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.mlp	Gemma4TextMLP	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.32.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.32.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.32.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.32.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.32.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.32.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.32.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.32.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.32.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.33	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.33.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.33.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.33.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.33.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.33.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.33.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.33.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.33.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.33.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.33.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.34	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.34.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.34.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.34.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.34.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.34.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.34.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.34.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.34.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.34.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.34.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.34.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.34.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.34.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.34.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.34.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.per_layer_model_projection.weight	Linear	8960 × 1536	bfloat16	13,762,560
model.language_model.per_layer_projection_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.vision_tower	Gemma4VisionModel	–	–	–
model.vision_tower.encoder	Gemma4VisionEncoder	–	–	–
model.vision_tower.encoder.layers.0	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.0.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.0.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.0.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.0.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.0.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.0.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.0.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.0.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.0.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.0.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.0.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.0.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.0.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.0.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.0.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.1	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.1.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.1.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.1.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.1.mlp.gate_proj	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.1.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.1.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.1.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.1.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.1.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.1.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.1.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.1.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.1.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.1.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.1.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.1.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.2	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.2.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.2.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.2.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.2.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.2.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.2.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.2.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.2.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.2.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.2.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.2.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.2.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.2.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.2.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.2.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.3	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.3.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.3.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.3.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.3.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.3.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.3.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.3.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.3.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.3.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.3.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.3.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.3.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.3.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.3.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.3.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.4	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.4.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.4.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.4.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.4.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.4.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.4.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.4.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.4.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.4.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.4.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.4.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.4.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.4.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.4.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.4.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.5	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.5.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.5.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.5.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.5.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.5.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.5.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.5.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.5.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.5.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.5.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.5.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.5.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.5.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.5.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.5.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.6	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.6.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.6.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.6.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.6.mlp.gate_proj	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.6.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.6.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.6.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.6.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.6.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.6.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.6.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.6.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.6.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.6.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.6.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.6.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.7	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.7.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.7.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.7.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.7.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.7.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.7.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.7.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.7.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.7.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.7.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.7.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.7.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.7.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.7.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.7.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.8	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.8.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.8.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.8.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.8.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.8.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.8.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.8.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.8.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.8.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.8.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.8.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.8.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.8.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.8.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.8.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.9	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.9.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.9.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.9.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.9.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.9.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.9.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.9.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.9.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.9.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.9.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.9.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.9.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.9.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.9.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.9.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.10	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.10.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.10.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.10.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.10.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.10.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.10.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.10.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.10.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.10.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.10.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.10.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.10.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.10.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.10.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.10.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.11	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.11.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.11.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.11.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.11.mlp.gate_proj	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.11.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.11.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.11.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.11.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.11.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.11.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.11.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.11.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.11.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.11.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.11.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.11.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.12	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.12.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.12.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.12.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.12.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.12.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.12.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.12.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.12.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.12.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.12.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.12.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.12.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.12.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.12.self_attn.q_proj	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.12.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.12.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.13	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.13.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.13.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.13.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.13.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.13.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.13.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.13.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.13.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.13.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.13.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.13.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.13.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.13.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.13.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.13.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.14	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.14.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.14.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.14.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.14.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.14.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.14.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.14.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.14.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.14.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.14.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.14.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.14.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.14.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.14.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.14.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.15	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.15.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.15.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.15.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.15.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.15.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.15.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.15.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.15.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.15.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.15.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.15.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.15.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.15.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.15.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.15.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.patch_embedder	Gemma4VisionPatchEmbedder	–	–	–
model.vision_tower.patch_embedder.input_proj.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.patch_embedder.position_embedding_table	Gemma4VisionPatchEmbedder	2 × 10240 × 768	bfloat16	15,728,640

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.audio_tower.layers.0.feed_forward1.ffw_layer_1.linear	4,194,304	4,194,304
model.audio_tower.layers.0.feed_forward1.ffw_layer_2.linear	4,194,304	8,388,608
model.audio_tower.layers.0.feed_forward1.post_layer_norm	1,024	8,389,632
model.audio_tower.layers.0.feed_forward1.pre_layer_norm	1,024	8,390,656
model.audio_tower.layers.0.feed_forward2.ffw_layer_1.linear	4,194,304	12,584,960
model.audio_tower.layers.0.feed_forward2.ffw_layer_2.linear	4,194,304	16,779,264
model.audio_tower.layers.0.feed_forward2.post_layer_norm	1,024	16,780,288
model.audio_tower.layers.0.feed_forward2.pre_layer_norm	1,024	16,781,312
model.audio_tower.layers.0.lconv1d.conv_norm	1,024	16,782,336
model.audio_tower.layers.0.lconv1d.depthwise_conv1d	5,120	16,787,456
model.audio_tower.layers.0.lconv1d.linear_end.linear	1,048,576	17,836,032
model.audio_tower.layers.0.lconv1d.linear_start.linear	2,097,152	19,933,184
model.audio_tower.layers.0.lconv1d.pre_layer_norm	1,024	19,934,208
model.audio_tower.layers.0.norm_out	1,024	19,935,232
model.audio_tower.layers.0.norm_post_attn	1,024	19,936,256
model.audio_tower.layers.0.norm_pre_attn	1,024	19,937,280
model.audio_tower.layers.0.self_attn.k_proj.linear	1,048,576	20,985,856
model.audio_tower.layers.0.self_attn	128	20,985,984
model.audio_tower.layers.0.self_attn.post.linear	1,048,576	22,034,560
model.audio_tower.layers.0.self_attn.q_proj.linear	1,048,576	23,083,136
model.audio_tower.layers.0.self_attn.relative_k_proj	1,048,576	24,131,712
model.audio_tower.layers.0.self_attn.v_proj.linear	1,048,576	25,180,288
model.audio_tower.layers.1.feed_forward1.ffw_layer_1.linear	4,194,304	29,374,592
model.audio_tower.layers.1.feed_forward1.ffw_layer_2.linear	4,194,304	33,568,896
model.audio_tower.layers.1.feed_forward1.post_layer_norm	1,024	33,569,920
model.audio_tower.layers.1.feed_forward1.pre_layer_norm	1,024	33,570,944
model.audio_tower.layers.1.feed_forward2.ffw_layer_1.linear	4,194,304	37,765,248
model.audio_tower.layers.1.feed_forward2.ffw_layer_2.linear	4,194,304	41,959,552
model.audio_tower.layers.1.feed_forward2.post_layer_norm	1,024	41,960,576
model.audio_tower.layers.1.feed_forward2.pre_layer_norm	1,024	41,961,600
model.audio_tower.layers.1.lconv1d.conv_norm	1,024	41,962,624
model.audio_tower.layers.1.lconv1d.depthwise_conv1d	5,120	41,967,744
model.audio_tower.layers.1.lconv1d.linear_end.linear	1,048,576	43,016,320
model.audio_tower.layers.1.lconv1d.linear_start.linear	2,097,152	45,113,472
model.audio_tower.layers.1.lconv1d.pre_layer_norm	1,024	45,114,496

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Module	Params	Running total
model.audio_tower.layers.1.norm_out	1,024	45,115,520
model.audio_tower.layers.1.norm_post_attn	1,024	45,116,544
model.audio_tower.layers.1.norm_pre_attn	1,024	45,117,568
model.audio_tower.layers.1.self_attn.k_proj.linear	1,048,576	46,166,144
model.audio_tower.layers.1.self_attn	128	46,166,272
model.audio_tower.layers.1.self_attn.post.linear	1,048,576	47,214,848
model.audio_tower.layers.1.self_attn.q_proj.linear	1,048,576	48,263,424
model.audio_tower.layers.1.self_attn. relative_k_proj	1,048,576	49,312,000
model.audio_tower.layers.1.self_attn.v_proj.linear	1,048,576	50,360,576
model.audio_tower.layers.2.feed_forward1. ffw_layer_1.linear	4,194,304	54,554,880
model.audio_tower.layers.2.feed_forward1. ffw_layer_2.linear	4,194,304	58,749,184
model.audio_tower.layers.2.feed_forward1. post_layer_norm	1,024	58,750,208
model.audio_tower.layers.2.feed_forward1. pre_layer_norm	1,024	58,751,232
model.audio_tower.layers.2.feed_forward2. ffw_layer_1.linear	4,194,304	62,945,536
model.audio_tower.layers.2.feed_forward2. ffw_layer_2.linear	4,194,304	67,139,840
model.audio_tower.layers.2.feed_forward2. post_layer_norm	1,024	67,140,864
model.audio_tower.layers.2.feed_forward2. pre_layer_norm	1,024	67,141,888
model.audio_tower.layers.2.lconv1d.conv_norm	1,024	67,142,912
model.audio_tower.layers.2.lconv1d.depthwise_conv1d	5,120	67,148,032
model.audio_tower.layers.2.lconv1d.linear_end. linear	1,048,576	68,196,608
model.audio_tower.layers.2.lconv1d.linear_start. linear	2,097,152	70,293,760
model.audio_tower.layers.2.lconv1d.pre_layer_norm	1,024	70,294,784
model.audio_tower.layers.2.norm_out	1,024	70,295,808
model.audio_tower.layers.2.norm_post_attn	1,024	70,296,832
model.audio_tower.layers.2.norm_pre_attn	1,024	70,297,856
model.audio_tower.layers.2.self_attn.k_proj.linear	1,048,576	71,346,432
model.audio_tower.layers.2.self_attn	128	71,346,560
model.audio_tower.layers.2.self_attn.post.linear	1,048,576	72,395,136
model.audio_tower.layers.2.self_attn.q_proj.linear	1,048,576	73,443,712
model.audio_tower.layers.2.self_attn. relative_k_proj	1,048,576	74,492,288
model.audio_tower.layers.2.self_attn.v_proj.linear	1,048,576	75,540,864
model.audio_tower.layers.3.feed_forward1. ffw_layer_1.linear	4,194,304	79,735,168
model.audio_tower.layers.3.feed_forward1. ffw_layer_2.linear	4,194,304	83,929,472
model.audio_tower.layers.3.feed_forward1. post_layer_norm	1,024	83,930,496
model.audio_tower.layers.3.feed_forward1. pre_layer_norm	1,024	83,931,520
model.audio_tower.layers.3.feed_forward2. ffw_layer_1.linear	4,194,304	88,125,824
model.audio_tower.layers.3.feed_forward2. ffw_layer_2.linear	4,194,304	92,320,128
model.audio_tower.layers.3.feed_forward2. post_layer_norm	1,024	92,321,152

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Module	Params	Running total
model.audio_tower.layers.3.feed_forward2.pre_layer_norm	1,024	92,322,176
model.audio_tower.layers.3.lconv1d.conv_norm	1,024	92,323,200
model.audio_tower.layers.3.lconv1d.depthwise_conv1d	5,120	92,328,320
model.audio_tower.layers.3.lconv1d.linear_end.linear	1,048,576	93,376,896
model.audio_tower.layers.3.lconv1d.linear_start.linear	2,097,152	95,474,048
model.audio_tower.layers.3.lconv1d.pre_layer_norm	1,024	95,475,072
model.audio_tower.layers.3.norm_out	1,024	95,476,096
model.audio_tower.layers.3.norm_post_attn	1,024	95,477,120
model.audio_tower.layers.3.norm_pre_attn	1,024	95,478,144
model.audio_tower.layers.3.self_attn.k_proj.linear	1,048,576	96,526,720
model.audio_tower.layers.3.self_attn	128	96,526,848
model.audio_tower.layers.3.self_attn.post.linear	1,048,576	97,575,424
model.audio_tower.layers.3.self_attn.q_proj.linear	1,048,576	98,624,000
model.audio_tower.layers.3.self_attn.relative_k_proj	1,048,576	99,672,576
model.audio_tower.layers.3.self_attn.v_proj.linear	1,048,576	100,721,152
model.audio_tower.layers.4.feed_forward1.ffw_layer_1.linear	4,194,304	104,915,456
model.audio_tower.layers.4.feed_forward1.ffw_layer_2.linear	4,194,304	109,109,760
model.audio_tower.layers.4.feed_forward1.post_layer_norm	1,024	109,110,784
model.audio_tower.layers.4.feed_forward1.pre_layer_norm	1,024	109,111,808
model.audio_tower.layers.4.feed_forward2.ffw_layer_1.linear	4,194,304	113,306,112
model.audio_tower.layers.4.feed_forward2.ffw_layer_2.linear	4,194,304	117,500,416
model.audio_tower.layers.4.feed_forward2.post_layer_norm	1,024	117,501,440
model.audio_tower.layers.4.feed_forward2.pre_layer_norm	1,024	117,502,464
model.audio_tower.layers.4.lconv1d.conv_norm	1,024	117,503,488
model.audio_tower.layers.4.lconv1d.depthwise_conv1d	5,120	117,508,608
model.audio_tower.layers.4.lconv1d.linear_end.linear	1,048,576	118,557,184
model.audio_tower.layers.4.lconv1d.linear_start.linear	2,097,152	120,654,336
model.audio_tower.layers.4.lconv1d.pre_layer_norm	1,024	120,655,360
model.audio_tower.layers.4.norm_out	1,024	120,656,384
model.audio_tower.layers.4.norm_post_attn	1,024	120,657,408
model.audio_tower.layers.4.norm_pre_attn	1,024	120,658,432
model.audio_tower.layers.4.self_attn.k_proj.linear	1,048,576	121,707,008
model.audio_tower.layers.4.self_attn	128	121,707,136
model.audio_tower.layers.4.self_attn.post.linear	1,048,576	122,755,712
model.audio_tower.layers.4.self_attn.q_proj.linear	1,048,576	123,804,288
model.audio_tower.layers.4.self_attn.relative_k_proj	1,048,576	124,852,864
model.audio_tower.layers.4.self_attn.v_proj.linear	1,048,576	125,901,440
model.audio_tower.layers.5.feed_forward1.ffw_layer_1.linear	4,194,304	130,095,744
model.audio_tower.layers.5.feed_forward1.ffw_layer_2.linear	4,194,304	134,290,048
model.audio_tower.layers.5.feed_forward1.post_layer_norm	1,024	134,291,072

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Module	Params	Running total
model.audio_tower.layers.5.feed_forward1.pre_layer_norm	1,024	134,292,096
model.audio_tower.layers.5.feed_forward2.ffw_layer_1.linear	4,194,304	138,486,400
model.audio_tower.layers.5.feed_forward2.ffw_layer_2.linear	4,194,304	142,680,704
model.audio_tower.layers.5.feed_forward2.post_layer_norm	1,024	142,681,728
model.audio_tower.layers.5.feed_forward2.pre_layer_norm	1,024	142,682,752
model.audio_tower.layers.5.lconv1d.conv_norm	1,024	142,683,776
model.audio_tower.layers.5.lconv1d.depthwise_conv1d	5,120	142,688,896
model.audio_tower.layers.5.lconv1d.linear_end.linear	1,048,576	143,737,472
model.audio_tower.layers.5.lconv1d.linear_start.linear	2,097,152	145,834,624
model.audio_tower.layers.5.lconv1d.pre_layer_norm	1,024	145,835,648
model.audio_tower.layers.5.norm_out	1,024	145,836,672
model.audio_tower.layers.5.norm_post_attn	1,024	145,837,696
model.audio_tower.layers.5.norm_pre_attn	1,024	145,838,720
model.audio_tower.layers.5.self_attn.k_proj.linear	1,048,576	146,887,296
model.audio_tower.layers.5.self_attn	128	146,887,424
model.audio_tower.layers.5.self_attn.post.linear	1,048,576	147,936,000
model.audio_tower.layers.5.self_attn.q_proj.linear	1,048,576	148,984,576
model.audio_tower.layers.5.self_attn.relative_k_proj	1,048,576	150,033,152
model.audio_tower.layers.5.self_attn.v_proj.linear	1,048,576	151,081,728
model.audio_tower.layers.6.feed_forward1.ffw_layer_1.linear	4,194,304	155,276,032
model.audio_tower.layers.6.feed_forward1.ffw_layer_2.linear	4,194,304	159,470,336
model.audio_tower.layers.6.feed_forward1.post_layer_norm	1,024	159,471,360
model.audio_tower.layers.6.feed_forward1.pre_layer_norm	1,024	159,472,384
model.audio_tower.layers.6.feed_forward2.ffw_layer_1.linear	4,194,304	163,666,688
model.audio_tower.layers.6.feed_forward2.ffw_layer_2.linear	4,194,304	167,860,992
model.audio_tower.layers.6.feed_forward2.post_layer_norm	1,024	167,862,016
model.audio_tower.layers.6.feed_forward2.pre_layer_norm	1,024	167,863,040
model.audio_tower.layers.6.lconv1d.conv_norm	1,024	167,864,064
model.audio_tower.layers.6.lconv1d.depthwise_conv1d	5,120	167,869,184
model.audio_tower.layers.6.lconv1d.linear_end.linear	1,048,576	168,917,760
model.audio_tower.layers.6.lconv1d.linear_start.linear	2,097,152	171,014,912
model.audio_tower.layers.6.lconv1d.pre_layer_norm	1,024	171,015,936
model.audio_tower.layers.6.norm_out	1,024	171,016,960
model.audio_tower.layers.6.norm_post_attn	1,024	171,017,984
model.audio_tower.layers.6.norm_pre_attn	1,024	171,019,008
model.audio_tower.layers.6.self_attn.k_proj.linear	1,048,576	172,067,584
model.audio_tower.layers.6.self_attn	128	172,067,712
model.audio_tower.layers.6.self_attn.post.linear	1,048,576	173,116,288
model.audio_tower.layers.6.self_attn.q_proj.linear	1,048,576	174,164,864

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Module	Params	Running total
model.audio_tower.layers.6.self_attn.relative_k_proj	1,048,576	175,213,440
model.audio_tower.layers.6.self_attn.v_proj.linear	1,048,576	176,262,016
model.audio_tower.layers.7.feed_forward1.ffw_layer_1.linear	4,194,304	180,456,320
model.audio_tower.layers.7.feed_forward1.ffw_layer_2.linear	4,194,304	184,650,624
model.audio_tower.layers.7.feed_forward1.post_layer_norm	1,024	184,651,648
model.audio_tower.layers.7.feed_forward1.pre_layer_norm	1,024	184,652,672
model.audio_tower.layers.7.feed_forward2.ffw_layer_1.linear	4,194,304	188,846,976
model.audio_tower.layers.7.feed_forward2.ffw_layer_2.linear	4,194,304	193,041,280
model.audio_tower.layers.7.feed_forward2.post_layer_norm	1,024	193,042,304
model.audio_tower.layers.7.feed_forward2.pre_layer_norm	1,024	193,043,328
model.audio_tower.layers.7.lconv1d.conv_norm	1,024	193,044,352
model.audio_tower.layers.7.lconv1d.depthwise_conv1d	5,120	193,049,472
model.audio_tower.layers.7.lconv1d.linear_end.linear	1,048,576	194,098,048
model.audio_tower.layers.7.lconv1d.linear_start.linear	2,097,152	196,195,200
model.audio_tower.layers.7.lconv1d.pre_layer_norm	1,024	196,196,224
model.audio_tower.layers.7.norm_out	1,024	196,197,248
model.audio_tower.layers.7.norm_post_attn	1,024	196,198,272
model.audio_tower.layers.7.norm_pre_attn	1,024	196,199,296
model.audio_tower.layers.7.self_attn.k_proj.linear	1,048,576	197,247,872
model.audio_tower.layers.7.self_attn	128	197,248,000
model.audio_tower.layers.7.self_attn.post.linear	1,048,576	198,296,576
model.audio_tower.layers.7.self_attn.q_proj.linear	1,048,576	199,345,152
model.audio_tower.layers.7.self_attn.relative_k_proj	1,048,576	200,393,728
model.audio_tower.layers.7.self_attn.v_proj.linear	1,048,576	201,442,304
model.audio_tower.layers.8.feed_forward1.ffw_layer_1.linear	4,194,304	205,636,608
model.audio_tower.layers.8.feed_forward1.ffw_layer_2.linear	4,194,304	209,830,912
model.audio_tower.layers.8.feed_forward1.post_layer_norm	1,024	209,831,936
model.audio_tower.layers.8.feed_forward1.pre_layer_norm	1,024	209,832,960
model.audio_tower.layers.8.feed_forward2.ffw_layer_1.linear	4,194,304	214,027,264
model.audio_tower.layers.8.feed_forward2.ffw_layer_2.linear	4,194,304	218,221,568
model.audio_tower.layers.8.feed_forward2.post_layer_norm	1,024	218,222,592
model.audio_tower.layers.8.feed_forward2.pre_layer_norm	1,024	218,223,616
model.audio_tower.layers.8.lconv1d.conv_norm	1,024	218,224,640
model.audio_tower.layers.8.lconv1d.depthwise_conv1d	5,120	218,229,760
model.audio_tower.layers.8.lconv1d.linear_end.linear	1,048,576	219,278,336
model.audio_tower.layers.8.lconv1d.linear_start.linear	2,097,152	221,375,488

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Module	Params	Running total
model.audio_tower.layers.8.lconv1d.pre_layer_norm	1,024	221,376,512
model.audio_tower.layers.8.norm_out	1,024	221,377,536
model.audio_tower.layers.8.norm_post_attn	1,024	221,378,560
model.audio_tower.layers.8.norm_pre_attn	1,024	221,379,584
model.audio_tower.layers.8.self_attn.k_proj.linear	1,048,576	222,428,160
model.audio_tower.layers.8.self_attn	128	222,428,288
model.audio_tower.layers.8.self_attn.post.linear	1,048,576	223,476,864
model.audio_tower.layers.8.self_attn.q_proj.linear	1,048,576	224,525,440
model.audio_tower.layers.8.self_attn. relative_k_proj	1,048,576	225,574,016
model.audio_tower.layers.8.self_attn.v_proj.linear	1,048,576	226,622,592
model.audio_tower.layers.9.feed_forward1. ffw_layer_1.linear	4,194,304	230,816,896
model.audio_tower.layers.9.feed_forward1. ffw_layer_2.linear	4,194,304	235,011,200
model.audio_tower.layers.9.feed_forward1. post_layer_norm	1,024	235,012,224
model.audio_tower.layers.9.feed_forward1. pre_layer_norm	1,024	235,013,248
model.audio_tower.layers.9.feed_forward2. ffw_layer_1.linear	4,194,304	239,207,552
model.audio_tower.layers.9.feed_forward2. ffw_layer_2.linear	4,194,304	243,401,856
model.audio_tower.layers.9.feed_forward2. post_layer_norm	1,024	243,402,880
model.audio_tower.layers.9.feed_forward2. pre_layer_norm	1,024	243,403,904
model.audio_tower.layers.9.lconv1d.conv_norm	1,024	243,404,928
model.audio_tower.layers.9.lconv1d.depthwise_conv1d	5,120	243,410,048
model.audio_tower.layers.9.lconv1d.linear_end. linear	1,048,576	244,458,624
model.audio_tower.layers.9.lconv1d.linear_start. linear	2,097,152	246,555,776
model.audio_tower.layers.9.lconv1d.pre_layer_norm	1,024	246,556,800
model.audio_tower.layers.9.norm_out	1,024	246,557,824
model.audio_tower.layers.9.norm_post_attn	1,024	246,558,848
model.audio_tower.layers.9.norm_pre_attn	1,024	246,559,872
model.audio_tower.layers.9.self_attn.k_proj.linear	1,048,576	247,608,448
model.audio_tower.layers.9.self_attn	128	247,608,576
model.audio_tower.layers.9.self_attn.post.linear	1,048,576	248,657,152
model.audio_tower.layers.9.self_attn.q_proj.linear	1,048,576	249,705,728
model.audio_tower.layers.9.self_attn. relative_k_proj	1,048,576	250,754,304
model.audio_tower.layers.9.self_attn.v_proj.linear	1,048,576	251,802,880
model.audio_tower.layers.10.feed_forward1. ffw_layer_1.linear	4,194,304	255,997,184
model.audio_tower.layers.10.feed_forward1. ffw_layer_2.linear	4,194,304	260,191,488
model.audio_tower.layers.10.feed_forward1. post_layer_norm	1,024	260,192,512
model.audio_tower.layers.10.feed_forward1. pre_layer_norm	1,024	260,193,536
model.audio_tower.layers.10.feed_forward2. ffw_layer_1.linear	4,194,304	264,387,840
model.audio_tower.layers.10.feed_forward2. ffw_layer_2.linear	4,194,304	268,582,144
model.audio_tower.layers.10.feed_forward2. post_layer_norm	1,024	268,583,168

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Module	Params	Running total
model.audio_tower.layers.10.feed_forward2.pre_layer_norm	1,024	268,584,192
model.audio_tower.layers.10.lconv1d.conv_norm	1,024	268,585,216
model.audio_tower.layers.10.lconv1d.depthwise_conv1d	5,120	268,590,336
model.audio_tower.layers.10.lconv1d.linear_end.linear	1,048,576	269,638,912
model.audio_tower.layers.10.lconv1d.linear_start.linear	2,097,152	271,736,064
model.audio_tower.layers.10.lconv1d.pre_layer_norm	1,024	271,737,088
model.audio_tower.layers.10.norm_out	1,024	271,738,112
model.audio_tower.layers.10.norm_post_attn	1,024	271,739,136
model.audio_tower.layers.10.norm_pre_attn	1,024	271,740,160
model.audio_tower.layers.10.self_attn.k_proj.linear	1,048,576	272,788,736
model.audio_tower.layers.10.self_attn	128	272,788,864
model.audio_tower.layers.10.self_attn.post.linear	1,048,576	273,837,440
model.audio_tower.layers.10.self_attn.q_proj.linear	1,048,576	274,886,016
model.audio_tower.layers.10.self_attn.relative_k_proj	1,048,576	275,934,592
model.audio_tower.layers.10.self_attn.v_proj.linear	1,048,576	276,983,168
model.audio_tower.layers.11.feed_forward1.ffw_layer_1.linear	4,194,304	281,177,472
model.audio_tower.layers.11.feed_forward1.ffw_layer_2.linear	4,194,304	285,371,776
model.audio_tower.layers.11.feed_forward1.post_layer_norm	1,024	285,372,800
model.audio_tower.layers.11.feed_forward1.pre_layer_norm	1,024	285,373,824
model.audio_tower.layers.11.feed_forward2.ffw_layer_1.linear	4,194,304	289,568,128
model.audio_tower.layers.11.feed_forward2.ffw_layer_2.linear	4,194,304	293,762,432
model.audio_tower.layers.11.feed_forward2.post_layer_norm	1,024	293,763,456
model.audio_tower.layers.11.feed_forward2.pre_layer_norm	1,024	293,764,480
model.audio_tower.layers.11.lconv1d.conv_norm	1,024	293,765,504
model.audio_tower.layers.11.lconv1d.depthwise_conv1d	5,120	293,770,624
model.audio_tower.layers.11.lconv1d.linear_end.linear	1,048,576	294,819,200
model.audio_tower.layers.11.lconv1d.linear_start.linear	2,097,152	296,916,352
model.audio_tower.layers.11.lconv1d.pre_layer_norm	1,024	296,917,376
model.audio_tower.layers.11.norm_out	1,024	296,918,400
model.audio_tower.layers.11.norm_post_attn	1,024	296,919,424
model.audio_tower.layers.11.norm_pre_attn	1,024	296,920,448
model.audio_tower.layers.11.self_attn.k_proj.linear	1,048,576	297,969,024
model.audio_tower.layers.11.self_attn	128	297,969,152
model.audio_tower.layers.11.self_attn.post.linear	1,048,576	299,017,728
model.audio_tower.layers.11.self_attn.q_proj.linear	1,048,576	300,066,304
model.audio_tower.layers.11.self_attn.relative_k_proj	1,048,576	301,114,880
model.audio_tower.layers.11.self_attn.v_proj.linear	1,048,576	302,163,456
model.audio_tower.output_proj	1,574,400	303,737,856
model.audio_tower.subsample_conv_projection.input_proj_linear	1,048,576	304,786,432

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Module	Params	Running total
model.audio_tower.subsample_conv_projection.layer0.conv	1,152	304,787,584
model.audio_tower.subsample_conv_projection.layer0.norm	128	304,787,712
model.audio_tower.subsample_conv_projection.layer1.conv	36,864	304,824,576
model.audio_tower.subsample_conv_projection.layer1.norm	32	304,824,608
model.embed_audio.embedding_projection	2,359,296	307,183,904
model.embed_vision.embedding_projection	1,179,648	308,363,552
model.language_model.embed_tokens	402,653,184	711,016,736
model.language_model.embed_tokens_per_layer	2,348,810,240	3,059,826,976
model.language_model.layers.0.input_layernorm	1,536	3,059,828,512
model.language_model.layers.0.mlp.down_proj	9,437,184	3,069,265,696
model.language_model.layers.0.mlp.gate_proj	9,437,184	3,078,702,880
model.language_model.layers.0.mlp.up_proj	9,437,184	3,088,140,064
model.language_model.layers.0.per_layer_input_gate	393,216	3,088,533,280
model.language_model.layers.0.per_layer_projection	393,216	3,088,926,496
model.language_model.layers.0.post_attention_layernorm	1,536	3,088,928,032
model.language_model.layers.0.post_feedforward_layernorm	1,536	3,088,929,568
model.language_model.layers.0.post_per_layer_input_norm	1,536	3,088,931,104
model.language_model.layers.0.pre_feedforward_layernorm	1,536	3,088,932,640
model.language_model.layers.0.self_attn.k_norm	256	3,088,932,896
model.language_model.layers.0.self_attn.k_proj	393,216	3,089,326,112
model.language_model.layers.0.self_attn.o_proj	3,145,728	3,092,471,840
model.language_model.layers.0.self_attn.q_norm	256	3,092,472,096
model.language_model.layers.0.self_attn.q_proj	3,145,728	3,095,617,824
model.language_model.layers.0.self_attn.v_proj	393,216	3,096,011,040
model.language_model.layers.1.input_layernorm	1,536	3,096,012,576
model.language_model.layers.1.mlp.down_proj	9,437,184	3,105,449,760
model.language_model.layers.1.mlp.gate_proj	9,437,184	3,114,886,944
model.language_model.layers.1.mlp.up_proj	9,437,184	3,124,324,128
model.language_model.layers.1.per_layer_input_gate	393,216	3,124,717,344
model.language_model.layers.1.per_layer_projection	393,216	3,125,110,560
model.language_model.layers.1.post_attention_layernorm	1,536	3,125,112,096
model.language_model.layers.1.post_feedforward_layernorm	1,536	3,125,113,632
model.language_model.layers.1.post_per_layer_input_norm	1,536	3,125,115,168
model.language_model.layers.1.pre_feedforward_layernorm	1,536	3,125,116,704
model.language_model.layers.1.self_attn.k_norm	256	3,125,116,960
model.language_model.layers.1.self_attn.k_proj	393,216	3,125,510,176
model.language_model.layers.1.self_attn.o_proj	3,145,728	3,128,655,904
model.language_model.layers.1.self_attn.q_norm	256	3,128,656,160
model.language_model.layers.1.self_attn.q_proj	3,145,728	3,131,801,888
model.language_model.layers.1.self_attn.v_proj	393,216	3,132,195,104
model.language_model.layers.2.input_layernorm	1,536	3,132,196,640
model.language_model.layers.2.mlp.down_proj	9,437,184	3,141,633,824
model.language_model.layers.2.mlp.gate_proj	9,437,184	3,151,071,008
model.language_model.layers.2.mlp.up_proj	9,437,184	3,160,508,192
model.language_model.layers.2.per_layer_input_gate	393,216	3,160,901,408
model.language_model.layers.2.per_layer_projection	393,216	3,161,294,624

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Module	Params	Running total
model.language_model.layers.2. post_attention_layernorm	1,536	3,161,296,160
model.language_model.layers.2. post_feedforward_layernorm	1,536	3,161,297,696
model.language_model.layers.2. post_per_layer_input_norm	1,536	3,161,299,232
model.language_model.layers.2. pre_feedforward_layernorm	1,536	3,161,300,768
model.language_model.layers.2.self_attn.k_norm	256	3,161,301,024
model.language_model.layers.2.self_attn.k_proj	393,216	3,161,694,240
model.language_model.layers.2.self_attn.o_proj	3,145,728	3,164,839,968
model.language_model.layers.2.self_attn.q_norm	256	3,164,840,224
model.language_model.layers.2.self_attn.q_proj	3,145,728	3,167,985,952
model.language_model.layers.2.self_attn.v_proj	393,216	3,168,379,168
model.language_model.layers.3.input_layernorm	1,536	3,168,380,704
model.language_model.layers.3.mlp.down_proj	9,437,184	3,177,817,888
model.language_model.layers.3.mlp.gate_proj	9,437,184	3,187,255,072
model.language_model.layers.3.mlp.up_proj	9,437,184	3,196,692,256
model.language_model.layers.3.per_layer_input_gate	393,216	3,197,085,472
model.language_model.layers.3.per_layer_projection	393,216	3,197,478,688
model.language_model.layers.3. post_attention_layernorm	1,536	3,197,480,224
model.language_model.layers.3. post_feedforward_layernorm	1,536	3,197,481,760
model.language_model.layers.3. post_per_layer_input_norm	1,536	3,197,483,296
model.language_model.layers.3. pre_feedforward_layernorm	1,536	3,197,484,832
model.language_model.layers.3.self_attn.k_norm	256	3,197,485,088
model.language_model.layers.3.self_attn.k_proj	393,216	3,197,878,304
model.language_model.layers.3.self_attn.o_proj	3,145,728	3,201,024,032
model.language_model.layers.3.self_attn.q_norm	256	3,201,024,288
model.language_model.layers.3.self_attn.q_proj	3,145,728	3,204,170,016
model.language_model.layers.3.self_attn.v_proj	393,216	3,204,563,232
model.language_model.layers.4.input_layernorm	1,536	3,204,564,768
model.language_model.layers.4.mlp.down_proj	9,437,184	3,214,001,952
model.language_model.layers.4.mlp.gate_proj	9,437,184	3,223,439,136
model.language_model.layers.4.mlp.up_proj	9,437,184	3,232,876,320
model.language_model.layers.4.per_layer_input_gate	393,216	3,233,269,536
model.language_model.layers.4.per_layer_projection	393,216	3,233,662,752
model.language_model.layers.4. post_attention_layernorm	1,536	3,233,664,288
model.language_model.layers.4. post_feedforward_layernorm	1,536	3,233,665,824
model.language_model.layers.4. post_per_layer_input_norm	1,536	3,233,667,360
model.language_model.layers.4. pre_feedforward_layernorm	1,536	3,233,668,896
model.language_model.layers.4.self_attn.k_norm	512	3,233,669,408
model.language_model.layers.4.self_attn.k_proj	786,432	3,234,455,840
model.language_model.layers.4.self_attn.o_proj	6,291,456	3,240,747,296
model.language_model.layers.4.self_attn.q_norm	512	3,240,747,808
model.language_model.layers.4.self_attn.q_proj	6,291,456	3,247,039,264
model.language_model.layers.4.self_attn.v_proj	786,432	3,247,825,696
model.language_model.layers.5.input_layernorm	1,536	3,247,827,232
model.language_model.layers.5.mlp.down_proj	9,437,184	3,257,264,416
model.language_model.layers.5.mlp.gate_proj	9,437,184	3,266,701,600
model.language_model.layers.5.mlp.up_proj	9,437,184	3,276,138,784

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Module	Params	Running total
model.language_model.layers.5.per_layer_input_gate	393,216	3,276,532,000
model.language_model.layers.5.per_layer_projection	393,216	3,276,925,216
model.language_model.layers.5. post_attention_layernorm	1,536	3,276,926,752
model.language_model.layers.5. post_feedforward_layernorm	1,536	3,276,928,288
model.language_model.layers.5. post_per_layer_input_norm	1,536	3,276,929,824
model.language_model.layers.5. pre_feedforward_layernorm	1,536	3,276,931,360
model.language_model.layers.5.self_attn.k_norm	256	3,276,931,616
model.language_model.layers.5.self_attn.k_proj	393,216	3,277,324,832
model.language_model.layers.5.self_attn.o_proj	3,145,728	3,280,470,560
model.language_model.layers.5.self_attn.q_norm	256	3,280,470,816
model.language_model.layers.5.self_attn.q_proj	3,145,728	3,283,616,544
model.language_model.layers.5.self_attn.v_proj	393,216	3,284,009,760
model.language_model.layers.6.input_layernorm	1,536	3,284,011,296
model.language_model.layers.6.mlp.down_proj	9,437,184	3,293,448,480
model.language_model.layers.6.mlp.gate_proj	9,437,184	3,302,885,664
model.language_model.layers.6.mlp.up_proj	9,437,184	3,312,322,848
model.language_model.layers.6.per_layer_input_gate	393,216	3,312,716,064
model.language_model.layers.6.per_layer_projection	393,216	3,313,109,280
model.language_model.layers.6. post_attention_layernorm	1,536	3,313,110,816
model.language_model.layers.6. post_feedforward_layernorm	1,536	3,313,112,352
model.language_model.layers.6. post_per_layer_input_norm	1,536	3,313,113,888
model.language_model.layers.6. pre_feedforward_layernorm	1,536	3,313,115,424
model.language_model.layers.6.self_attn.k_norm	256	3,313,115,680
model.language_model.layers.6.self_attn.k_proj	393,216	3,313,508,896
model.language_model.layers.6.self_attn.o_proj	3,145,728	3,316,654,624
model.language_model.layers.6.self_attn.q_norm	256	3,316,654,880
model.language_model.layers.6.self_attn.q_proj	3,145,728	3,319,800,608
model.language_model.layers.6.self_attn.v_proj	393,216	3,320,193,824
model.language_model.layers.7.input_layernorm	1,536	3,320,195,360
model.language_model.layers.7.mlp.down_proj	9,437,184	3,329,632,544
model.language_model.layers.7.mlp.gate_proj	9,437,184	3,339,069,728
model.language_model.layers.7.mlp.up_proj	9,437,184	3,348,506,912
model.language_model.layers.7.per_layer_input_gate	393,216	3,348,900,128
model.language_model.layers.7.per_layer_projection	393,216	3,349,293,344
model.language_model.layers.7. post_attention_layernorm	1,536	3,349,294,880
model.language_model.layers.7. post_feedforward_layernorm	1,536	3,349,296,416
model.language_model.layers.7. post_per_layer_input_norm	1,536	3,349,297,952
model.language_model.layers.7. pre_feedforward_layernorm	1,536	3,349,299,488
model.language_model.layers.7.self_attn.k_norm	256	3,349,299,744
model.language_model.layers.7.self_attn.k_proj	393,216	3,349,692,960
model.language_model.layers.7.self_attn.o_proj	3,145,728	3,352,838,688
model.language_model.layers.7.self_attn.q_norm	256	3,352,838,944
model.language_model.layers.7.self_attn.q_proj	3,145,728	3,355,984,672
model.language_model.layers.7.self_attn.v_proj	393,216	3,356,377,888
model.language_model.layers.8.input_layernorm	1,536	3,356,379,424
model.language_model.layers.8.mlp.down_proj	9,437,184	3,365,816,608

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Module	Params	Running total
model.language_model.layers.8.mlp.gate_proj	9,437,184	3,375,253,792
model.language_model.layers.8.mlp.up_proj	9,437,184	3,384,690,976
model.language_model.layers.8.per_layer_input_gate	393,216	3,385,084,192
model.language_model.layers.8.per_layer_projection	393,216	3,385,477,408
model.language_model.layers.8. post_attention_layernorm	1,536	3,385,478,944
model.language_model.layers.8. post_feedforward_layernorm	1,536	3,385,480,480
model.language_model.layers.8. post_per_layer_input_norm	1,536	3,385,482,016
model.language_model.layers.8. pre_feedforward_layernorm	1,536	3,385,483,552
model.language_model.layers.8.self_attn.k_norm	256	3,385,483,808
model.language_model.layers.8.self_attn.k_proj	393,216	3,385,877,024
model.language_model.layers.8.self_attn.o_proj	3,145,728	3,389,022,752
model.language_model.layers.8.self_attn.q_norm	256	3,389,023,008
model.language_model.layers.8.self_attn.q_proj	3,145,728	3,392,168,736
model.language_model.layers.8.self_attn.v_proj	393,216	3,392,561,952
model.language_model.layers.9.input_layernorm	1,536	3,392,563,488
model.language_model.layers.9.mlp.down_proj	9,437,184	3,402,000,672
model.language_model.layers.9.mlp.gate_proj	9,437,184	3,411,437,856
model.language_model.layers.9.mlp.up_proj	9,437,184	3,420,875,040
model.language_model.layers.9.per_layer_input_gate	393,216	3,421,268,256
model.language_model.layers.9.per_layer_projection	393,216	3,421,661,472
model.language_model.layers.9. post_attention_layernorm	1,536	3,421,663,008
model.language_model.layers.9. post_feedforward_layernorm	1,536	3,421,664,544
model.language_model.layers.9. post_per_layer_input_norm	1,536	3,421,666,080
model.language_model.layers.9. pre_feedforward_layernorm	1,536	3,421,667,616
model.language_model.layers.9.self_attn.k_norm	512	3,421,668,128
model.language_model.layers.9.self_attn.k_proj	786,432	3,422,454,560
model.language_model.layers.9.self_attn.o_proj	6,291,456	3,428,746,016
model.language_model.layers.9.self_attn.q_norm	512	3,428,746,528
model.language_model.layers.9.self_attn.q_proj	6,291,456	3,435,037,984
model.language_model.layers.9.self_attn.v_proj	786,432	3,435,824,416
model.language_model.layers.10.input_layernorm	1,536	3,435,825,952
model.language_model.layers.10.mlp.down_proj	9,437,184	3,445,263,136
model.language_model.layers.10.mlp.gate_proj	9,437,184	3,454,700,320
model.language_model.layers.10.mlp.up_proj	9,437,184	3,464,137,504
model.language_model.layers.10.per_layer_input_gate	393,216	3,464,530,720
model.language_model.layers.10.per_layer_projection	393,216	3,464,923,936
model.language_model.layers.10. post_attention_layernorm	1,536	3,464,925,472
model.language_model.layers.10. post_feedforward_layernorm	1,536	3,464,927,008
model.language_model.layers.10. post_per_layer_input_norm	1,536	3,464,928,544
model.language_model.layers.10. pre_feedforward_layernorm	1,536	3,464,930,080
model.language_model.layers.10.self_attn.k_norm	256	3,464,930,336
model.language_model.layers.10.self_attn.k_proj	393,216	3,465,323,552
model.language_model.layers.10.self_attn.o_proj	3,145,728	3,468,469,280
model.language_model.layers.10.self_attn.q_norm	256	3,468,469,536
model.language_model.layers.10.self_attn.q_proj	3,145,728	3,471,615,264
model.language_model.layers.10.self_attn.v_proj	393,216	3,472,008,480

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Module	Params	Running total
model.language_model.layers.11.input_layernorm	1,536	3,472,010,016
model.language_model.layers.11.mlp.down_proj	9,437,184	3,481,447,200
model.language_model.layers.11.mlp.gate_proj	9,437,184	3,490,884,384
model.language_model.layers.11.mlp.up_proj	9,437,184	3,500,321,568
model.language_model.layers.11.per_layer_input_gate	393,216	3,500,714,784
model.language_model.layers.11.per_layer_projection	393,216	3,501,108,000
model.language_model.layers.11.post_attention_layernorm	1,536	3,501,109,536
model.language_model.layers.11.post_feedforward_layernorm	1,536	3,501,111,072
model.language_model.layers.11.post_per_layer_input_norm	1,536	3,501,112,608
model.language_model.layers.11.pre_feedforward_layernorm	1,536	3,501,114,144
model.language_model.layers.11.self_attn.k_norm	256	3,501,114,400
model.language_model.layers.11.self_attn.k_proj	393,216	3,501,507,616
model.language_model.layers.11.self_attn.o_proj	3,145,728	3,504,653,344
model.language_model.layers.11.self_attn.q_norm	256	3,504,653,600
model.language_model.layers.11.self_attn.q_proj	3,145,728	3,507,799,328
model.language_model.layers.11.self_attn.v_proj	393,216	3,508,192,544
model.language_model.layers.12.input_layernorm	1,536	3,508,194,080
model.language_model.layers.12.mlp.down_proj	9,437,184	3,517,631,264
model.language_model.layers.12.mlp.gate_proj	9,437,184	3,527,068,448
model.language_model.layers.12.mlp.up_proj	9,437,184	3,536,505,632
model.language_model.layers.12.per_layer_input_gate	393,216	3,536,898,848
model.language_model.layers.12.per_layer_projection	393,216	3,537,292,064
model.language_model.layers.12.post_attention_layernorm	1,536	3,537,293,600
model.language_model.layers.12.post_feedforward_layernorm	1,536	3,537,295,136
model.language_model.layers.12.post_per_layer_input_norm	1,536	3,537,296,672
model.language_model.layers.12.pre_feedforward_layernorm	1,536	3,537,298,208
model.language_model.layers.12.self_attn.k_norm	256	3,537,298,464
model.language_model.layers.12.self_attn.k_proj	393,216	3,537,691,680
model.language_model.layers.12.self_attn.o_proj	3,145,728	3,540,837,408
model.language_model.layers.12.self_attn.q_norm	256	3,540,837,664
model.language_model.layers.12.self_attn.q_proj	3,145,728	3,543,983,392
model.language_model.layers.12.self_attn.v_proj	393,216	3,544,376,608
model.language_model.layers.13.input_layernorm	1,536	3,544,378,144
model.language_model.layers.13.mlp.down_proj	9,437,184	3,553,815,328
model.language_model.layers.13.mlp.gate_proj	9,437,184	3,563,252,512
model.language_model.layers.13.mlp.up_proj	9,437,184	3,572,689,696
model.language_model.layers.13.per_layer_input_gate	393,216	3,573,082,912
model.language_model.layers.13.per_layer_projection	393,216	3,573,476,128
model.language_model.layers.13.post_attention_layernorm	1,536	3,573,477,664
model.language_model.layers.13.post_feedforward_layernorm	1,536	3,573,479,200
model.language_model.layers.13.post_per_layer_input_norm	1,536	3,573,480,736
model.language_model.layers.13.pre_feedforward_layernorm	1,536	3,573,482,272
model.language_model.layers.13.self_attn.k_norm	256	3,573,482,528
model.language_model.layers.13.self_attn.k_proj	393,216	3,573,875,744
model.language_model.layers.13.self_attn.o_proj	3,145,728	3,577,021,472
model.language_model.layers.13.self_attn.q_norm	256	3,577,021,728

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Module	Params	Running total
model.language_model.layers.13.self_attn.q_proj	3,145,728	3,580,167,456
model.language_model.layers.13.self_attn.v_proj	393,216	3,580,560,672
model.language_model.layers.14.input_layernorm	1,536	3,580,562,208
model.language_model.layers.14.mlp.down_proj	9,437,184	3,589,999,392
model.language_model.layers.14.mlp.gate_proj	9,437,184	3,599,436,576
model.language_model.layers.14.mlp.up_proj	9,437,184	3,608,873,760
model.language_model.layers.14.per_layer_input_gate	393,216	3,609,266,976
model.language_model.layers.14.per_layer_projection	393,216	3,609,660,192
model.language_model.layers.14. post_attention_layernorm	1,536	3,609,661,728
model.language_model.layers.14. post_feedforward_layernorm	1,536	3,609,663,264
model.language_model.layers.14. post_per_layer_input_norm	1,536	3,609,664,800
model.language_model.layers.14. pre_feedforward_layernorm	1,536	3,609,666,336
model.language_model.layers.14.self_attn.k_norm	512	3,609,666,848
model.language_model.layers.14.self_attn.k_proj	786,432	3,610,453,280
model.language_model.layers.14.self_attn.o_proj	6,291,456	3,616,744,736
model.language_model.layers.14.self_attn.q_norm	512	3,616,745,248
model.language_model.layers.14.self_attn.q_proj	6,291,456	3,623,036,704
model.language_model.layers.14.self_attn.v_proj	786,432	3,623,823,136
model.language_model.layers.15.input_layernorm	1,536	3,623,824,672
model.language_model.layers.15.mlp.down_proj	18,874,368	3,642,699,040
model.language_model.layers.15.mlp.gate_proj	18,874,368	3,661,573,408
model.language_model.layers.15.mlp.up_proj	18,874,368	3,680,447,776
model.language_model.layers.15.per_layer_input_gate	393,216	3,680,840,992
model.language_model.layers.15.per_layer_projection	393,216	3,681,234,208
model.language_model.layers.15. post_attention_layernorm	1,536	3,681,235,744
model.language_model.layers.15. post_feedforward_layernorm	1,536	3,681,237,280
model.language_model.layers.15. post_per_layer_input_norm	1,536	3,681,238,816
model.language_model.layers.15. pre_feedforward_layernorm	1,536	3,681,240,352
model.language_model.layers.15.self_attn.o_proj	3,145,728	3,684,386,080
model.language_model.layers.15.self_attn.q_norm	256	3,684,386,336
model.language_model.layers.15.self_attn.q_proj	3,145,728	3,687,532,064
model.language_model.layers.16.input_layernorm	1,536	3,687,533,600
model.language_model.layers.16.mlp.down_proj	18,874,368	3,706,407,968
model.language_model.layers.16.mlp.gate_proj	18,874,368	3,725,282,336
model.language_model.layers.16.mlp.up_proj	18,874,368	3,744,156,704
model.language_model.layers.16.per_layer_input_gate	393,216	3,744,549,920
model.language_model.layers.16.per_layer_projection	393,216	3,744,943,136
model.language_model.layers.16. post_attention_layernorm	1,536	3,744,944,672
model.language_model.layers.16. post_feedforward_layernorm	1,536	3,744,946,208
model.language_model.layers.16. post_per_layer_input_norm	1,536	3,744,947,744
model.language_model.layers.16. pre_feedforward_layernorm	1,536	3,744,949,280
model.language_model.layers.16.self_attn.o_proj	3,145,728	3,748,095,008
model.language_model.layers.16.self_attn.q_norm	256	3,748,095,264
model.language_model.layers.16.self_attn.q_proj	3,145,728	3,751,240,992
model.language_model.layers.17.input_layernorm	1,536	3,751,242,528
model.language_model.layers.17.mlp.down_proj	18,874,368	3,770,116,896

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Module	Params	Running total
model.language_model.layers.17.mlp.gate_proj	18,874,368	3,788,991,264
model.language_model.layers.17.mlp.up_proj	18,874,368	3,807,865,632
model.language_model.layers.17.per_layer_input_gate	393,216	3,808,258,848
model.language_model.layers.17.per_layer_projection	393,216	3,808,652,064
model.language_model.layers.17. post_attention_layernorm	1,536	3,808,653,600
model.language_model.layers.17. post_feedforward_layernorm	1,536	3,808,655,136
model.language_model.layers.17. post_per_layer_input_norm	1,536	3,808,656,672
model.language_model.layers.17. pre_feedforward_layernorm	1,536	3,808,658,208
model.language_model.layers.17.self_attn.o_proj	3,145,728	3,811,803,936
model.language_model.layers.17.self_attn.q_norm	256	3,811,804,192
model.language_model.layers.17.self_attn.q_proj	3,145,728	3,814,949,920
model.language_model.layers.18.input_layernorm	1,536	3,814,951,456
model.language_model.layers.18.mlp.down_proj	18,874,368	3,833,825,824
model.language_model.layers.18.mlp.gate_proj	18,874,368	3,852,700,192
model.language_model.layers.18.mlp.up_proj	18,874,368	3,871,574,560
model.language_model.layers.18.per_layer_input_gate	393,216	3,871,967,776
model.language_model.layers.18.per_layer_projection	393,216	3,872,360,992
model.language_model.layers.18. post_attention_layernorm	1,536	3,872,362,528
model.language_model.layers.18. post_feedforward_layernorm	1,536	3,872,364,064
model.language_model.layers.18. post_per_layer_input_norm	1,536	3,872,365,600
model.language_model.layers.18. pre_feedforward_layernorm	1,536	3,872,367,136
model.language_model.layers.18.self_attn.o_proj	3,145,728	3,875,512,864
model.language_model.layers.18.self_attn.q_norm	256	3,875,513,120
model.language_model.layers.18.self_attn.q_proj	3,145,728	3,878,658,848
model.language_model.layers.19.input_layernorm	1,536	3,878,660,384
model.language_model.layers.19.mlp.down_proj	18,874,368	3,897,534,752
model.language_model.layers.19.mlp.gate_proj	18,874,368	3,916,409,120
model.language_model.layers.19.mlp.up_proj	18,874,368	3,935,283,488
model.language_model.layers.19.per_layer_input_gate	393,216	3,935,676,704
model.language_model.layers.19.per_layer_projection	393,216	3,936,069,920
model.language_model.layers.19. post_attention_layernorm	1,536	3,936,071,456
model.language_model.layers.19. post_feedforward_layernorm	1,536	3,936,072,992
model.language_model.layers.19. post_per_layer_input_norm	1,536	3,936,074,528
model.language_model.layers.19. pre_feedforward_layernorm	1,536	3,936,076,064
model.language_model.layers.19.self_attn.o_proj	6,291,456	3,942,367,520
model.language_model.layers.19.self_attn.q_norm	512	3,942,368,032
model.language_model.layers.19.self_attn.q_proj	6,291,456	3,948,659,488
model.language_model.layers.20.input_layernorm	1,536	3,948,661,024
model.language_model.layers.20.mlp.down_proj	18,874,368	3,967,535,392
model.language_model.layers.20.mlp.gate_proj	18,874,368	3,986,409,760
model.language_model.layers.20.mlp.up_proj	18,874,368	4,005,284,128
model.language_model.layers.20.per_layer_input_gate	393,216	4,005,677,344
model.language_model.layers.20.per_layer_projection	393,216	4,006,070,560
model.language_model.layers.20. post_attention_layernorm	1,536	4,006,072,096

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Module	Params	Running total
model.language_model.layers.20.post_feedforward_layernorm	1,536	4,006,073,632
model.language_model.layers.20.post_per_layer_input_norm	1,536	4,006,075,168
model.language_model.layers.20.pre_feedforward_layernorm	1,536	4,006,076,704
model.language_model.layers.20.self_attn.o_proj	3,145,728	4,009,222,432
model.language_model.layers.20.self_attn.q_norm	256	4,009,222,688
model.language_model.layers.20.self_attn.q_proj	3,145,728	4,012,368,416
model.language_model.layers.21.input_layernorm	1,536	4,012,369,952
model.language_model.layers.21.mlp.down_proj	18,874,368	4,031,244,320
model.language_model.layers.21.mlp.gate_proj	18,874,368	4,050,118,688
model.language_model.layers.21.mlp.up_proj	18,874,368	4,068,993,056
model.language_model.layers.21.per_layer_input_gate	393,216	4,069,386,272
model.language_model.layers.21.per_layer_projection	393,216	4,069,779,488
model.language_model.layers.21.post_attention_layernorm	1,536	4,069,781,024
model.language_model.layers.21.post_feedforward_layernorm	1,536	4,069,782,560
model.language_model.layers.21.post_per_layer_input_norm	1,536	4,069,784,096
model.language_model.layers.21.pre_feedforward_layernorm	1,536	4,069,785,632
model.language_model.layers.21.self_attn.o_proj	3,145,728	4,072,931,360
model.language_model.layers.21.self_attn.q_norm	256	4,072,931,616
model.language_model.layers.21.self_attn.q_proj	3,145,728	4,076,077,344
model.language_model.layers.22.input_layernorm	1,536	4,076,078,880
model.language_model.layers.22.mlp.down_proj	18,874,368	4,094,953,248
model.language_model.layers.22.mlp.gate_proj	18,874,368	4,113,827,616
model.language_model.layers.22.mlp.up_proj	18,874,368	4,132,701,984
model.language_model.layers.22.per_layer_input_gate	393,216	4,133,095,200
model.language_model.layers.22.per_layer_projection	393,216	4,133,488,416
model.language_model.layers.22.post_attention_layernorm	1,536	4,133,489,952
model.language_model.layers.22.post_feedforward_layernorm	1,536	4,133,491,488
model.language_model.layers.22.post_per_layer_input_norm	1,536	4,133,493,024
model.language_model.layers.22.pre_feedforward_layernorm	1,536	4,133,494,560
model.language_model.layers.22.self_attn.o_proj	3,145,728	4,136,640,288
model.language_model.layers.22.self_attn.q_norm	256	4,136,640,544
model.language_model.layers.22.self_attn.q_proj	3,145,728	4,139,786,272
model.language_model.layers.23.input_layernorm	1,536	4,139,787,808
model.language_model.layers.23.mlp.down_proj	18,874,368	4,158,662,176
model.language_model.layers.23.mlp.gate_proj	18,874,368	4,177,536,544
model.language_model.layers.23.mlp.up_proj	18,874,368	4,196,410,912
model.language_model.layers.23.per_layer_input_gate	393,216	4,196,804,128
model.language_model.layers.23.per_layer_projection	393,216	4,197,197,344
model.language_model.layers.23.post_attention_layernorm	1,536	4,197,198,880
model.language_model.layers.23.post_feedforward_layernorm	1,536	4,197,200,416
model.language_model.layers.23.post_per_layer_input_norm	1,536	4,197,201,952
model.language_model.layers.23.pre_feedforward_layernorm	1,536	4,197,203,488
model.language_model.layers.23.self_attn.o_proj	3,145,728	4,200,349,216

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Module	Params	Running total
model.language_model.layers.23.self_attn.q_norm	256	4,200,349,472
model.language_model.layers.23.self_attn.q_proj	3,145,728	4,203,495,200
model.language_model.layers.24.input_layernorm	1,536	4,203,496,736
model.language_model.layers.24.mlp.down_proj	18,874,368	4,222,371,104
model.language_model.layers.24.mlp.gate_proj	18,874,368	4,241,245,472
model.language_model.layers.24.mlp.up_proj	18,874,368	4,260,119,840
model.language_model.layers.24.per_layer_input_gate	393,216	4,260,513,056
model.language_model.layers.24.per_layer_projection	393,216	4,260,906,272
model.language_model.layers.24.post_attention_layernorm	1,536	4,260,907,808
model.language_model.layers.24.post_feedforward_layernorm	1,536	4,260,909,344
model.language_model.layers.24.post_per_layer_input_norm	1,536	4,260,910,880
model.language_model.layers.24.pre_feedforward_layernorm	1,536	4,260,912,416
model.language_model.layers.24.self_attn.o_proj	6,291,456	4,267,203,872
model.language_model.layers.24.self_attn.q_norm	512	4,267,204,384
model.language_model.layers.24.self_attn.q_proj	6,291,456	4,273,495,840
model.language_model.layers.25.input_layernorm	1,536	4,273,497,376
model.language_model.layers.25.mlp.down_proj	18,874,368	4,292,371,744
model.language_model.layers.25.mlp.gate_proj	18,874,368	4,311,246,112
model.language_model.layers.25.mlp.up_proj	18,874,368	4,330,120,480
model.language_model.layers.25.per_layer_input_gate	393,216	4,330,513,696
model.language_model.layers.25.per_layer_projection	393,216	4,330,906,912
model.language_model.layers.25.post_attention_layernorm	1,536	4,330,908,448
model.language_model.layers.25.post_feedforward_layernorm	1,536	4,330,909,984
model.language_model.layers.25.post_per_layer_input_norm	1,536	4,330,911,520
model.language_model.layers.25.pre_feedforward_layernorm	1,536	4,330,913,056
model.language_model.layers.25.self_attn.o_proj	3,145,728	4,334,058,784
model.language_model.layers.25.self_attn.q_norm	256	4,334,059,040
model.language_model.layers.25.self_attn.q_proj	3,145,728	4,337,204,768
model.language_model.layers.26.input_layernorm	1,536	4,337,206,304
model.language_model.layers.26.mlp.down_proj	18,874,368	4,356,080,672
model.language_model.layers.26.mlp.gate_proj	18,874,368	4,374,955,040
model.language_model.layers.26.mlp.up_proj	18,874,368	4,393,829,408
model.language_model.layers.26.per_layer_input_gate	393,216	4,394,222,624
model.language_model.layers.26.per_layer_projection	393,216	4,394,615,840
model.language_model.layers.26.post_attention_layernorm	1,536	4,394,617,376
model.language_model.layers.26.post_feedforward_layernorm	1,536	4,394,618,912
model.language_model.layers.26.post_per_layer_input_norm	1,536	4,394,620,448
model.language_model.layers.26.pre_feedforward_layernorm	1,536	4,394,621,984
model.language_model.layers.26.self_attn.o_proj	3,145,728	4,397,767,712
model.language_model.layers.26.self_attn.q_norm	256	4,397,767,968
model.language_model.layers.26.self_attn.q_proj	3,145,728	4,400,913,696
model.language_model.layers.27.input_layernorm	1,536	4,400,915,232
model.language_model.layers.27.mlp.down_proj	18,874,368	4,419,789,600
model.language_model.layers.27.mlp.gate_proj	18,874,368	4,438,663,968
model.language_model.layers.27.mlp.up_proj	18,874,368	4,457,538,336
model.language_model.layers.27.per_layer_input_gate	393,216	4,457,931,552

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Module	Params	Running total
model.language_model.layers.27.per_layer_projection	393,216	4,458,324,768
model.language_model.layers.27.post_attention_layernorm	1,536	4,458,326,304
model.language_model.layers.27.post_feedforward_layernorm	1,536	4,458,327,840
model.language_model.layers.27.post_per_layer_input_norm	1,536	4,458,329,376
model.language_model.layers.27.pre_feedforward_layernorm	1,536	4,458,330,912
model.language_model.layers.27.self_attn.o_proj	3,145,728	4,461,476,640
model.language_model.layers.27.self_attn.q_norm	256	4,461,476,896
model.language_model.layers.27.self_attn.q_proj	3,145,728	4,464,622,624
model.language_model.layers.28.input_layernorm	1,536	4,464,624,160
model.language_model.layers.28.mlp.down_proj	18,874,368	4,483,498,528
model.language_model.layers.28.mlp.gate_proj	18,874,368	4,502,372,896
model.language_model.layers.28.mlp.up_proj	18,874,368	4,521,247,264
model.language_model.layers.28.per_layer_input_gate	393,216	4,521,640,480
model.language_model.layers.28.per_layer_projection	393,216	4,522,033,696
model.language_model.layers.28.post_attention_layernorm	1,536	4,522,035,232
model.language_model.layers.28.post_feedforward_layernorm	1,536	4,522,036,768
model.language_model.layers.28.post_per_layer_input_norm	1,536	4,522,038,304
model.language_model.layers.28.pre_feedforward_layernorm	1,536	4,522,039,840
model.language_model.layers.28.self_attn.o_proj	3,145,728	4,525,185,568
model.language_model.layers.28.self_attn.q_norm	256	4,525,185,824
model.language_model.layers.28.self_attn.q_proj	3,145,728	4,528,331,552
model.language_model.layers.29.input_layernorm	1,536	4,528,333,088
model.language_model.layers.29.mlp.down_proj	18,874,368	4,547,207,456
model.language_model.layers.29.mlp.gate_proj	18,874,368	4,566,081,824
model.language_model.layers.29.mlp.up_proj	18,874,368	4,584,956,192
model.language_model.layers.29.per_layer_input_gate	393,216	4,585,349,408
model.language_model.layers.29.per_layer_projection	393,216	4,585,742,624
model.language_model.layers.29.post_attention_layernorm	1,536	4,585,744,160
model.language_model.layers.29.post_feedforward_layernorm	1,536	4,585,745,696
model.language_model.layers.29.post_per_layer_input_norm	1,536	4,585,747,232
model.language_model.layers.29.pre_feedforward_layernorm	1,536	4,585,748,768
model.language_model.layers.29.self_attn.o_proj	6,291,456	4,592,040,224
model.language_model.layers.29.self_attn.q_norm	512	4,592,040,736
model.language_model.layers.29.self_attn.q_proj	6,291,456	4,598,332,192
model.language_model.layers.30.input_layernorm	1,536	4,598,333,728
model.language_model.layers.30.mlp.down_proj	18,874,368	4,617,208,096
model.language_model.layers.30.mlp.gate_proj	18,874,368	4,636,082,464
model.language_model.layers.30.mlp.up_proj	18,874,368	4,654,956,832
model.language_model.layers.30.per_layer_input_gate	393,216	4,655,350,048
model.language_model.layers.30.per_layer_projection	393,216	4,655,743,264
model.language_model.layers.30.post_attention_layernorm	1,536	4,655,744,800
model.language_model.layers.30.post_feedforward_layernorm	1,536	4,655,746,336
model.language_model.layers.30.post_per_layer_input_norm	1,536	4,655,747,872

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Module	Params	Running total
model.language_model.layers.30. pre_feedforward_layernorm	1,536	4,655,749,408
model.language_model.layers.30.self_attn.o_proj	3,145,728	4,658,895,136
model.language_model.layers.30.self_attn.q_norm	256	4,658,895,392
model.language_model.layers.30.self_attn.q_proj	3,145,728	4,662,041,120
model.language_model.layers.31.input_layernorm	1,536	4,662,042,656
model.language_model.layers.31.mlp.down_proj	18,874,368	4,680,917,024
model.language_model.layers.31.mlp.gate_proj	18,874,368	4,699,791,392
model.language_model.layers.31.mlp.up_proj	18,874,368	4,718,665,760
model.language_model.layers.31.per_layer_input_gate	393,216	4,719,058,976
model.language_model.layers.31.per_layer_projection	393,216	4,719,452,192
model.language_model.layers.31. post_attention_layernorm	1,536	4,719,453,728
model.language_model.layers.31. post_feedforward_layernorm	1,536	4,719,455,264
model.language_model.layers.31. post_per_layer_input_norm	1,536	4,719,456,800
model.language_model.layers.31. pre_feedforward_layernorm	1,536	4,719,458,336
model.language_model.layers.31.self_attn.o_proj	3,145,728	4,722,604,064
model.language_model.layers.31.self_attn.q_norm	256	4,722,604,320
model.language_model.layers.31.self_attn.q_proj	3,145,728	4,725,750,048
model.language_model.layers.32.input_layernorm	1,536	4,725,751,584
model.language_model.layers.32.mlp.down_proj	18,874,368	4,744,625,952
model.language_model.layers.32.mlp.gate_proj	18,874,368	4,763,500,320
model.language_model.layers.32.mlp.up_proj	18,874,368	4,782,374,688
model.language_model.layers.32.per_layer_input_gate	393,216	4,782,767,904
model.language_model.layers.32.per_layer_projection	393,216	4,783,161,120
model.language_model.layers.32. post_attention_layernorm	1,536	4,783,162,656
model.language_model.layers.32. post_feedforward_layernorm	1,536	4,783,164,192
model.language_model.layers.32. post_per_layer_input_norm	1,536	4,783,165,728
model.language_model.layers.32. pre_feedforward_layernorm	1,536	4,783,167,264
model.language_model.layers.32.self_attn.o_proj	3,145,728	4,786,312,992
model.language_model.layers.32.self_attn.q_norm	256	4,786,313,248
model.language_model.layers.32.self_attn.q_proj	3,145,728	4,789,458,976
model.language_model.layers.33.input_layernorm	1,536	4,789,460,512
model.language_model.layers.33.mlp.down_proj	18,874,368	4,808,334,880
model.language_model.layers.33.mlp.gate_proj	18,874,368	4,827,209,248
model.language_model.layers.33.mlp.up_proj	18,874,368	4,846,083,616
model.language_model.layers.33.per_layer_input_gate	393,216	4,846,476,832
model.language_model.layers.33.per_layer_projection	393,216	4,846,870,048
model.language_model.layers.33. post_attention_layernorm	1,536	4,846,871,584
model.language_model.layers.33. post_feedforward_layernorm	1,536	4,846,873,120
model.language_model.layers.33. post_per_layer_input_norm	1,536	4,846,874,656
model.language_model.layers.33. pre_feedforward_layernorm	1,536	4,846,876,192
model.language_model.layers.33.self_attn.o_proj	3,145,728	4,850,021,920
model.language_model.layers.33.self_attn.q_norm	256	4,850,022,176
model.language_model.layers.33.self_attn.q_proj	3,145,728	4,853,167,904
model.language_model.layers.34.input_layernorm	1,536	4,853,169,440
model.language_model.layers.34.mlp.down_proj	18,874,368	4,872,043,808

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Module	Params	Running total
model.language_model.layers.34.mlp.gate_proj	18,874,368	4,890,918,176
model.language_model.layers.34.mlp.up_proj	18,874,368	4,909,792,544
model.language_model.layers.34.per_layer_input_gate	393,216	4,910,185,760
model.language_model.layers.34.per_layer_projection	393,216	4,910,578,976
model.language_model.layers.34. post_attention_layernorm	1,536	4,910,580,512
model.language_model.layers.34. post_feedforward_layernorm	1,536	4,910,582,048
model.language_model.layers.34. post_per_layer_input_norm	1,536	4,910,583,584
model.language_model.layers.34. pre_feedforward_layernorm	1,536	4,910,585,120
model.language_model.layers.34.self_attn.o_proj	6,291,456	4,916,876,576
model.language_model.layers.34.self_attn.q_norm	512	4,916,877,088
model.language_model.layers.34.self_attn.q_proj	6,291,456	4,923,168,544
model.language_model.norm	1,536	4,923,170,080
model.language_model.per_layer_model_projection	13,762,560	4,936,932,640
model.language_model.per_layer_projection_norm	256	4,936,932,896
model.vision_tower.encoder.layers.0.input_layernorm	768	4,936,933,664
model.vision_tower.encoder.layers.0.mlp.down_proj. linear	2,359,296	4,939,292,960
model.vision_tower.encoder.layers.0.mlp.gate_proj. linear	2,359,296	4,941,652,256
model.vision_tower.encoder.layers.0.mlp.up_proj. linear	2,359,296	4,944,011,552
model.vision_tower.encoder.layers.0. post_attention_layernorm	768	4,944,012,320
model.vision_tower.encoder.layers.0. post_feedforward_layernorm	768	4,944,013,088
model.vision_tower.encoder.layers.0. pre_feedforward_layernorm	768	4,944,013,856
model.vision_tower.encoder.layers.0.self_attn. k_norm	64	4,944,013,920
model.vision_tower.encoder.layers.0.self_attn. k_proj.linear	589,824	4,944,603,744
model.vision_tower.encoder.layers.0.self_attn. o_proj.linear	589,824	4,945,193,568
model.vision_tower.encoder.layers.0.self_attn. q_norm	64	4,945,193,632
model.vision_tower.encoder.layers.0.self_attn. q_proj.linear	589,824	4,945,783,456
model.vision_tower.encoder.layers.0.self_attn. v_proj.linear	589,824	4,946,373,280
model.vision_tower.encoder.layers.1.input_layernorm	768	4,946,374,048
model.vision_tower.encoder.layers.1.mlp.down_proj. linear	2,359,296	4,948,733,344
model.vision_tower.encoder.layers.1.mlp.gate_proj. linear	2,359,296	4,951,092,640
model.vision_tower.encoder.layers.1.mlp.up_proj. linear	2,359,296	4,953,451,936
model.vision_tower.encoder.layers.1. post_attention_layernorm	768	4,953,452,704
model.vision_tower.encoder.layers.1. post_feedforward_layernorm	768	4,953,453,472
model.vision_tower.encoder.layers.1. pre_feedforward_layernorm	768	4,953,454,240
model.vision_tower.encoder.layers.1.self_attn. k_norm	64	4,953,454,304

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Module	Params	Running total
model.vision_tower.encoder.layers.1.self_attn.k_proj.linear	589,824	4,954,044,128
model.vision_tower.encoder.layers.1.self_attn.o_proj.linear	589,824	4,954,633,952
model.vision_tower.encoder.layers.1.self_attn.q_norm	64	4,954,634,016
model.vision_tower.encoder.layers.1.self_attn.q_proj.linear	589,824	4,955,223,840
model.vision_tower.encoder.layers.1.self_attn.v_proj.linear	589,824	4,955,813,664
model.vision_tower.encoder.layers.2.input_layernorm	768	4,955,814,432
model.vision_tower.encoder.layers.2.mlp.down_proj.linear	2,359,296	4,958,173,728
model.vision_tower.encoder.layers.2.mlp.gate_proj.linear	2,359,296	4,960,533,024
model.vision_tower.encoder.layers.2.mlp.up_proj.linear	2,359,296	4,962,892,320
model.vision_tower.encoder.layers.2.post_attention_layernorm	768	4,962,893,088
model.vision_tower.encoder.layers.2.post_feedforward_layernorm	768	4,962,893,856
model.vision_tower.encoder.layers.2.pre_feedforward_layernorm	768	4,962,894,624
model.vision_tower.encoder.layers.2.self_attn.k_norm	64	4,962,894,688
model.vision_tower.encoder.layers.2.self_attn.k_proj.linear	589,824	4,963,484,512
model.vision_tower.encoder.layers.2.self_attn.o_proj.linear	589,824	4,964,074,336
model.vision_tower.encoder.layers.2.self_attn.q_norm	64	4,964,074,400
model.vision_tower.encoder.layers.2.self_attn.q_proj.linear	589,824	4,964,664,224
model.vision_tower.encoder.layers.2.self_attn.v_proj.linear	589,824	4,965,254,048
model.vision_tower.encoder.layers.3.input_layernorm	768	4,965,254,816
model.vision_tower.encoder.layers.3.mlp.down_proj.linear	2,359,296	4,967,614,112
model.vision_tower.encoder.layers.3.mlp.gate_proj.linear	2,359,296	4,969,973,408
model.vision_tower.encoder.layers.3.mlp.up_proj.linear	2,359,296	4,972,332,704
model.vision_tower.encoder.layers.3.post_attention_layernorm	768	4,972,333,472
model.vision_tower.encoder.layers.3.post_feedforward_layernorm	768	4,972,334,240
model.vision_tower.encoder.layers.3.pre_feedforward_layernorm	768	4,972,335,008
model.vision_tower.encoder.layers.3.self_attn.k_norm	64	4,972,335,072
model.vision_tower.encoder.layers.3.self_attn.k_proj.linear	589,824	4,972,924,896
model.vision_tower.encoder.layers.3.self_attn.o_proj.linear	589,824	4,973,514,720
model.vision_tower.encoder.layers.3.self_attn.q_norm	64	4,973,514,784
model.vision_tower.encoder.layers.3.self_attn.q_proj.linear	589,824	4,974,104,608

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Module	Params	Running total
model.vision_tower.encoder.layers.3.self_attn.v_proj.linear	589,824	4,974,694,432
model.vision_tower.encoder.layers.4.input_layernorm	768	4,974,695,200
model.vision_tower.encoder.layers.4.mlp.down_proj.linear	2,359,296	4,977,054,496
model.vision_tower.encoder.layers.4.mlp.gate_proj.linear	2,359,296	4,979,413,792
model.vision_tower.encoder.layers.4.mlp.up_proj.linear	2,359,296	4,981,773,088
model.vision_tower.encoder.layers.4.post_attention_layernorm	768	4,981,773,856
model.vision_tower.encoder.layers.4.post_feedforward_layernorm	768	4,981,774,624
model.vision_tower.encoder.layers.4.pre_feedforward_layernorm	768	4,981,775,392
model.vision_tower.encoder.layers.4.self_attn.k_norm	64	4,981,775,456
model.vision_tower.encoder.layers.4.self_attn.k_proj.linear	589,824	4,982,365,280
model.vision_tower.encoder.layers.4.self_attn.o_proj.linear	589,824	4,982,955,104
model.vision_tower.encoder.layers.4.self_attn.q_norm	64	4,982,955,168
model.vision_tower.encoder.layers.4.self_attn.q_proj.linear	589,824	4,983,544,992
model.vision_tower.encoder.layers.4.self_attn.v_proj.linear	589,824	4,984,134,816
model.vision_tower.encoder.layers.5.input_layernorm	768	4,984,135,584
model.vision_tower.encoder.layers.5.mlp.down_proj.linear	2,359,296	4,986,494,880
model.vision_tower.encoder.layers.5.mlp.gate_proj.linear	2,359,296	4,988,854,176
model.vision_tower.encoder.layers.5.mlp.up_proj.linear	2,359,296	4,991,213,472
model.vision_tower.encoder.layers.5.post_attention_layernorm	768	4,991,214,240
model.vision_tower.encoder.layers.5.post_feedforward_layernorm	768	4,991,215,008
model.vision_tower.encoder.layers.5.pre_feedforward_layernorm	768	4,991,215,776
model.vision_tower.encoder.layers.5.self_attn.k_norm	64	4,991,215,840
model.vision_tower.encoder.layers.5.self_attn.k_proj.linear	589,824	4,991,805,664
model.vision_tower.encoder.layers.5.self_attn.o_proj.linear	589,824	4,992,395,488
model.vision_tower.encoder.layers.5.self_attn.q_norm	64	4,992,395,552
model.vision_tower.encoder.layers.5.self_attn.q_proj.linear	589,824	4,992,985,376
model.vision_tower.encoder.layers.5.self_attn.v_proj.linear	589,824	4,993,575,200
model.vision_tower.encoder.layers.6.input_layernorm	768	4,993,575,968
model.vision_tower.encoder.layers.6.mlp.down_proj.linear	2,359,296	4,995,935,264
model.vision_tower.encoder.layers.6.mlp.gate_proj.linear	2,359,296	4,998,294,560

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Module	Params	Running total
model.vision_tower.encoder.layers.6.mlp.up_proj.linear	2,359,296	5,000,653,856
model.vision_tower.encoder.layers.6.post_attention_layernorm	768	5,000,654,624
model.vision_tower.encoder.layers.6.post_feedforward_layernorm	768	5,000,655,392
model.vision_tower.encoder.layers.6.pre_feedforward_layernorm	768	5,000,656,160
model.vision_tower.encoder.layers.6.self_attn.k_norm	64	5,000,656,224
model.vision_tower.encoder.layers.6.self_attn.k_proj.linear	589,824	5,001,246,048
model.vision_tower.encoder.layers.6.self_attn.o_proj.linear	589,824	5,001,835,872
model.vision_tower.encoder.layers.6.self_attn.q_norm	64	5,001,835,936
model.vision_tower.encoder.layers.6.self_attn.q_proj.linear	589,824	5,002,425,760
model.vision_tower.encoder.layers.6.self_attn.v_proj.linear	589,824	5,003,015,584
model.vision_tower.encoder.layers.7.input_layernorm	768	5,003,016,352
model.vision_tower.encoder.layers.7.mlp.down_proj.linear	2,359,296	5,005,375,648
model.vision_tower.encoder.layers.7.mlp.gate_proj.linear	2,359,296	5,007,734,944
model.vision_tower.encoder.layers.7.mlp.up_proj.linear	2,359,296	5,010,094,240
model.vision_tower.encoder.layers.7.post_attention_layernorm	768	5,010,095,008
model.vision_tower.encoder.layers.7.post_feedforward_layernorm	768	5,010,095,776
model.vision_tower.encoder.layers.7.pre_feedforward_layernorm	768	5,010,096,544
model.vision_tower.encoder.layers.7.self_attn.k_norm	64	5,010,096,608
model.vision_tower.encoder.layers.7.self_attn.k_proj.linear	589,824	5,010,686,432
model.vision_tower.encoder.layers.7.self_attn.o_proj.linear	589,824	5,011,276,256
model.vision_tower.encoder.layers.7.self_attn.q_norm	64	5,011,276,320
model.vision_tower.encoder.layers.7.self_attn.q_proj.linear	589,824	5,011,866,144
model.vision_tower.encoder.layers.7.self_attn.v_proj.linear	589,824	5,012,455,968
model.vision_tower.encoder.layers.8.input_layernorm	768	5,012,456,736
model.vision_tower.encoder.layers.8.mlp.down_proj.linear	2,359,296	5,014,816,032
model.vision_tower.encoder.layers.8.mlp.gate_proj.linear	2,359,296	5,017,175,328
model.vision_tower.encoder.layers.8.mlp.up_proj.linear	2,359,296	5,019,534,624
model.vision_tower.encoder.layers.8.post_attention_layernorm	768	5,019,535,392
model.vision_tower.encoder.layers.8.post_feedforward_layernorm	768	5,019,536,160
model.vision_tower.encoder.layers.8.pre_feedforward_layernorm	768	5,019,536,928

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Module	Params	Running total
model.vision_tower.encoder.layers.8.self_attn.k_norm	64	5,019,536,992
model.vision_tower.encoder.layers.8.self_attn.k_proj.linear	589,824	5,020,126,816
model.vision_tower.encoder.layers.8.self_attn.o_proj.linear	589,824	5,020,716,640
model.vision_tower.encoder.layers.8.self_attn.q_norm	64	5,020,716,704
model.vision_tower.encoder.layers.8.self_attn.q_proj.linear	589,824	5,021,306,528
model.vision_tower.encoder.layers.8.self_attn.v_proj.linear	589,824	5,021,896,352
model.vision_tower.encoder.layers.9.input_layernorm	768	5,021,897,120
model.vision_tower.encoder.layers.9.mlp.down_proj.linear	2,359,296	5,024,256,416
model.vision_tower.encoder.layers.9.mlp.gate_proj.linear	2,359,296	5,026,615,712
model.vision_tower.encoder.layers.9.mlp.up_proj.linear	2,359,296	5,028,975,008
model.vision_tower.encoder.layers.9.post_attention_layernorm	768	5,028,975,776
model.vision_tower.encoder.layers.9.post_feedforward_layernorm	768	5,028,976,544
model.vision_tower.encoder.layers.9.pre_feedforward_layernorm	768	5,028,977,312
model.vision_tower.encoder.layers.9.self_attn.k_norm	64	5,028,977,376
model.vision_tower.encoder.layers.9.self_attn.k_proj.linear	589,824	5,029,567,200
model.vision_tower.encoder.layers.9.self_attn.o_proj.linear	589,824	5,030,157,024
model.vision_tower.encoder.layers.9.self_attn.q_norm	64	5,030,157,088
model.vision_tower.encoder.layers.9.self_attn.q_proj.linear	589,824	5,030,746,912
model.vision_tower.encoder.layers.9.self_attn.v_proj.linear	589,824	5,031,336,736
model.vision_tower.encoder.layers.10.input_layernorm	768	5,031,337,504
model.vision_tower.encoder.layers.10.mlp.down_proj.linear	2,359,296	5,033,696,800
model.vision_tower.encoder.layers.10.mlp.gate_proj.linear	2,359,296	5,036,056,096
model.vision_tower.encoder.layers.10.mlp.up_proj.linear	2,359,296	5,038,415,392
model.vision_tower.encoder.layers.10.post_attention_layernorm	768	5,038,416,160
model.vision_tower.encoder.layers.10.post_feedforward_layernorm	768	5,038,416,928
model.vision_tower.encoder.layers.10.pre_feedforward_layernorm	768	5,038,417,696
model.vision_tower.encoder.layers.10.self_attn.k_norm	64	5,038,417,760
model.vision_tower.encoder.layers.10.self_attn.k_proj.linear	589,824	5,039,007,584
model.vision_tower.encoder.layers.10.self_attn.o_proj.linear	589,824	5,039,597,408

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Module	Params	Running total
model.vision_tower.encoder.layers.10.self_attn.q_norm	64	5,039,597,472
model.vision_tower.encoder.layers.10.self_attn.q_proj.linear	589,824	5,040,187,296
model.vision_tower.encoder.layers.10.self_attn.v_proj.linear	589,824	5,040,777,120
model.vision_tower.encoder.layers.11.input_layernorm	768	5,040,777,888
model.vision_tower.encoder.layers.11.mlp.down_proj.linear	2,359,296	5,043,137,184
model.vision_tower.encoder.layers.11.mlp.gate_proj.linear	2,359,296	5,045,496,480
model.vision_tower.encoder.layers.11.mlp.up_proj.linear	2,359,296	5,047,855,776
model.vision_tower.encoder.layers.11.post_attention_layernorm	768	5,047,856,544
model.vision_tower.encoder.layers.11.post_feedforward_layernorm	768	5,047,857,312
model.vision_tower.encoder.layers.11.pre_feedforward_layernorm	768	5,047,858,080
model.vision_tower.encoder.layers.11.self_attn.k_norm	64	5,047,858,144
model.vision_tower.encoder.layers.11.self_attn.k_proj.linear	589,824	5,048,447,968
model.vision_tower.encoder.layers.11.self_attn.o_proj.linear	589,824	5,049,037,792
model.vision_tower.encoder.layers.11.self_attn.q_norm	64	5,049,037,856
model.vision_tower.encoder.layers.11.self_attn.q_proj.linear	589,824	5,049,627,680
model.vision_tower.encoder.layers.11.self_attn.v_proj.linear	589,824	5,050,217,504
model.vision_tower.encoder.layers.12.input_layernorm	768	5,050,218,272
model.vision_tower.encoder.layers.12.mlp.down_proj.linear	2,359,296	5,052,577,568
model.vision_tower.encoder.layers.12.mlp.gate_proj.linear	2,359,296	5,054,936,864
model.vision_tower.encoder.layers.12.mlp.up_proj.linear	2,359,296	5,057,296,160
model.vision_tower.encoder.layers.12.post_attention_layernorm	768	5,057,296,928
model.vision_tower.encoder.layers.12.post_feedforward_layernorm	768	5,057,297,696
model.vision_tower.encoder.layers.12.pre_feedforward_layernorm	768	5,057,298,464
model.vision_tower.encoder.layers.12.self_attn.k_norm	64	5,057,298,528
model.vision_tower.encoder.layers.12.self_attn.k_proj.linear	589,824	5,057,888,352
model.vision_tower.encoder.layers.12.self_attn.o_proj.linear	589,824	5,058,478,176
model.vision_tower.encoder.layers.12.self_attn.q_norm	64	5,058,478,240
model.vision_tower.encoder.layers.12.self_attn.q_proj.linear	589,824	5,059,068,064
model.vision_tower.encoder.layers.12.self_attn.v_proj.linear	589,824	5,059,657,888

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Module	Params	Running total
model.vision_tower.encoder.layers.13.input_layernorm	768	5,059,658,656
model.vision_tower.encoder.layers.13.mlp.down_proj.linear	2,359,296	5,062,017,952
model.vision_tower.encoder.layers.13.mlp.gate_proj.linear	2,359,296	5,064,377,248
model.vision_tower.encoder.layers.13.mlp.up_proj.linear	2,359,296	5,066,736,544
model.vision_tower.encoder.layers.13.post_attention_layernorm	768	5,066,737,312
model.vision_tower.encoder.layers.13.post_feedforward_layernorm	768	5,066,738,080
model.vision_tower.encoder.layers.13.pre_feedforward_layernorm	768	5,066,738,848
model.vision_tower.encoder.layers.13.self_attn.k_norm	64	5,066,738,912
model.vision_tower.encoder.layers.13.self_attn.k_proj.linear	589,824	5,067,328,736
model.vision_tower.encoder.layers.13.self_attn.o_proj.linear	589,824	5,067,918,560
model.vision_tower.encoder.layers.13.self_attn.q_norm	64	5,067,918,624
model.vision_tower.encoder.layers.13.self_attn.q_proj.linear	589,824	5,068,508,448
model.vision_tower.encoder.layers.13.self_attn.v_proj.linear	589,824	5,069,098,272
model.vision_tower.encoder.layers.14.input_layernorm	768	5,069,099,040
model.vision_tower.encoder.layers.14.mlp.down_proj.linear	2,359,296	5,071,458,336
model.vision_tower.encoder.layers.14.mlp.gate_proj.linear	2,359,296	5,073,817,632
model.vision_tower.encoder.layers.14.mlp.up_proj.linear	2,359,296	5,076,176,928
model.vision_tower.encoder.layers.14.post_attention_layernorm	768	5,076,177,696
model.vision_tower.encoder.layers.14.post_feedforward_layernorm	768	5,076,178,464
model.vision_tower.encoder.layers.14.pre_feedforward_layernorm	768	5,076,179,232
model.vision_tower.encoder.layers.14.self_attn.k_norm	64	5,076,179,296
model.vision_tower.encoder.layers.14.self_attn.k_proj.linear	589,824	5,076,769,120
model.vision_tower.encoder.layers.14.self_attn.o_proj.linear	589,824	5,077,358,944
model.vision_tower.encoder.layers.14.self_attn.q_norm	64	5,077,359,008
model.vision_tower.encoder.layers.14.self_attn.q_proj.linear	589,824	5,077,948,832
model.vision_tower.encoder.layers.14.self_attn.v_proj.linear	589,824	5,078,538,656
model.vision_tower.encoder.layers.15.input_layernorm	768	5,078,539,424
model.vision_tower.encoder.layers.15.mlp.down_proj.linear	2,359,296	5,080,898,720
model.vision_tower.encoder.layers.15.mlp.gate_proj.linear	2,359,296	5,083,258,016

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Module	Params	Running total
model.vision_tower.encoder.layers.15.mlp.up_proj.linear	2,359,296	5,085,617,312
model.vision_tower.encoder.layers.15.post_attention_layernorm	768	5,085,618,080
model.vision_tower.encoder.layers.15.post_feedforward_layernorm	768	5,085,618,848
model.vision_tower.encoder.layers.15.pre_feedforward_layernorm	768	5,085,619,616
model.vision_tower.encoder.layers.15.self_attn.k_norm	64	5,085,619,680
model.vision_tower.encoder.layers.15.self_attn.k_proj.linear	589,824	5,086,209,504
model.vision_tower.encoder.layers.15.self_attn.o_proj.linear	589,824	5,086,799,328
model.vision_tower.encoder.layers.15.self_attn.q_norm	64	5,086,799,392
model.vision_tower.encoder.layers.15.self_attn.q_proj.linear	589,824	5,087,389,216
model.vision_tower.encoder.layers.15.self_attn.v_proj.linear	589,824	5,087,979,040
model.vision_tower.patch_embedder.input_proj	589,824	5,088,568,864
model.vision_tower.patch_embedder	15,728,640	5,104,297,504